

Advanced Combustion

— Chapter 2. Thermodynamics and Transport Properties

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Outline of Chapter 2

- Introduction to Thermodynamics
- Fundamentals of Thermodynamics
- Thermodynamics in Combustion
- Transport Properties
- Transport in Combustion

Overview of Chapter 2

- To examine several thermodynamic concepts related to combustion and reacting systems: *element conservation, enthalpy, the first-law, adiabatic flame temperature, Chemical equilibrium, the second-law.*
- To provide fundamental conservation principles governing mass transfer and basic theories about heat and mass transfer in combustion.

Introduction

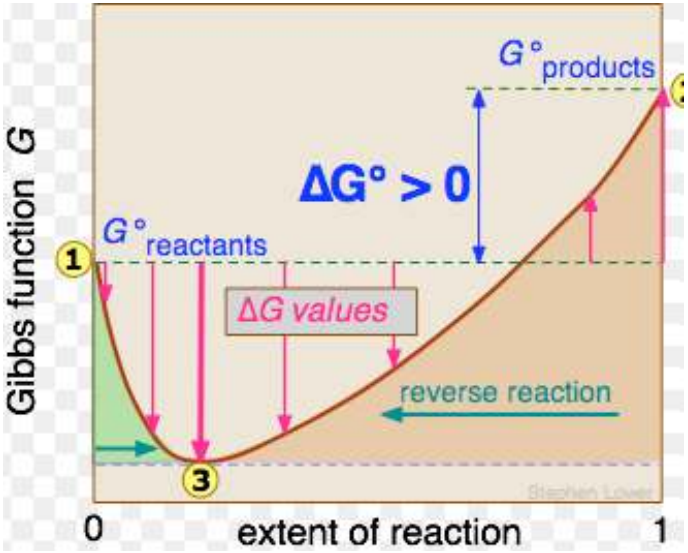
- *Thermodynamics* deals with equilibrium states and how chemical composition can be calculated for a system of known atomic/ molecular composition if two independent thermodynamic properties are known.
- Systems undergoing chemical reactions are not in general, in chemical equilibrium - reactions are rate-controlled.
- Chemical composition at a given instant in time is controlled by the thermodynamic properties, chemical reaction rates and fluid dynamics of the system.

Targets of Thermodynamics

We need thermodynamic data to:

- **Determine the heat release in a combustion process (need *enthalpies* and *heat capacities*).**
- **Calculate the equilibrium constant for a reaction – this allows us to relate the rate coefficients for *forward* and *reverse* reactions (need *enthalpies*, *entropies* and hence *Gibbs energies*, and *heat capacities*).**

Targets of Thermodynamics

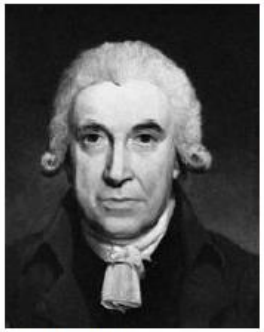




- To describe the driving force of a chemical process
- To determine the ideal end state of a reaction process
- Be the foundation of combustion

Giants of Thermodynamics

Watt steam engine (1765)



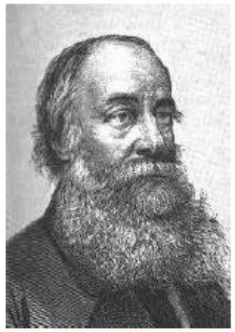
James Watt

Carnot cycle (1824)



Sati Carnot

Heat equivalent of work (1840)

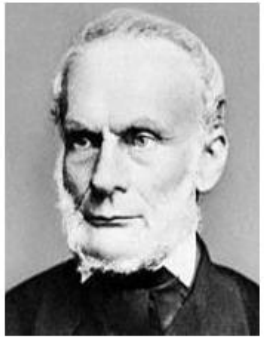


James Prescott Joule

Helmholtz free energy (1882)

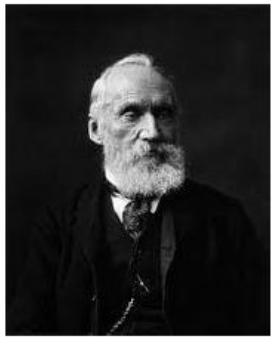


Hermann von Helmholtz



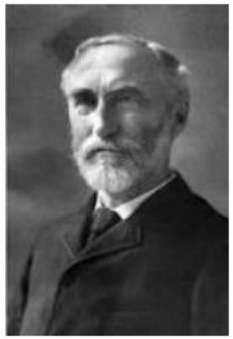
Rudolf J. E. Clausius

Second law of thermodynamics (1850)



William Thomson Kelvin

Kelvin scale (1848)



J Willard Gibbs

Chemical equilibrium (1876) Gibbs free energy (1878)

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Extensive and Intensive Properties

***Extensive property* (广延量):**

- Depends on the amount ("extent") of the matter. Summed up when identical systems are merged.
- e.g. amount of substance, volume, mass, denoted with capital letters, for example, V (m^3) for volume, U (J) for internal energy, H (J) (= U + PV) for enthalpy, etc.

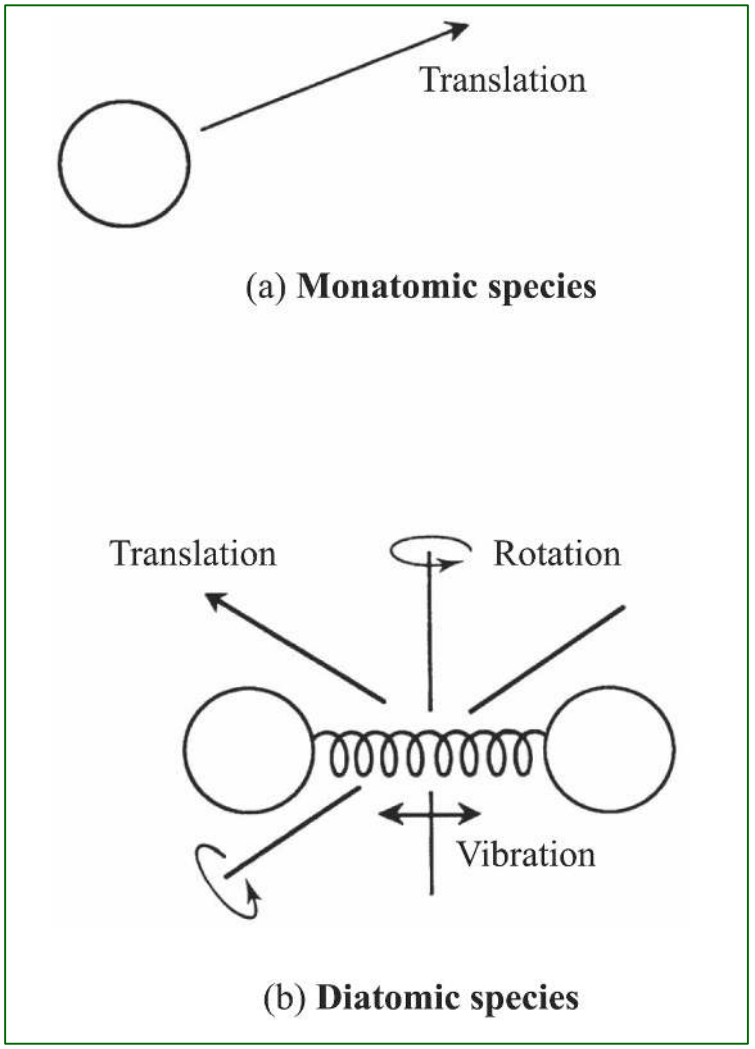
***Intensive property* (强度量):**

- Is expressed per unit mass (or per mole), hence independent of the amount of the matter. Levelled when systems are merged.
- e.g. temperature, pressure, density, denoted with lowercase letters, for example, v (m^3/kg) for specific volume, u (J/kg) for specific internal energy, h (J/kg) (= u + P*v*) for specific enthalpy, etc.

Internal Energy, Work and Heat

Internal energy:

- **The internal energy of an isolated system is a constant. Energy cannot be created or destroyed.**
- **Internal energy can have different forms:**
  - **Translational kinetic energy**
  - **Rotational energy**
  - **Vibrational energy**
  - **Energy within atoms**



Internal Energy, Work and Heat

Internal energy:

- **The internal energy U of a given state of the system is determined relative to that of a standard state of the system, by adding up the macroscopic transfers of energy that accompany a change of state from the reference state to the given state:** $\Delta U = \sum_i E_i$.

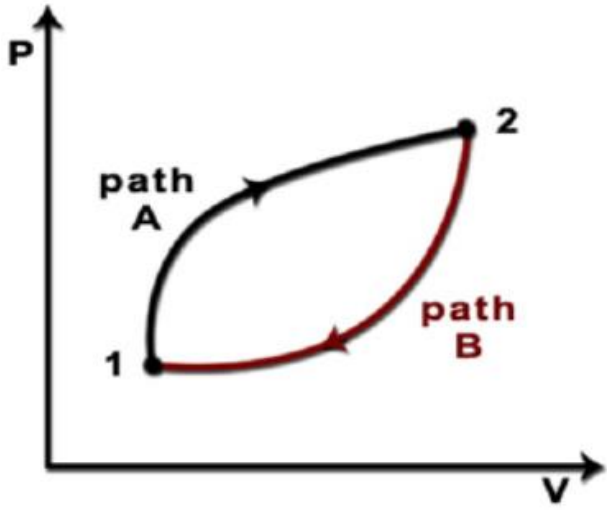
where ΔU denotes *the difference* between the internal energy of the given state and that of the reference state, and the E_i are the various energies transferred to the system in the steps from the reference state to the given state. It is the energy needed to create the given state of the system from the reference state.

Internal Energy, Work and Heat

Internal energy:

- Only the change of internal energy has physical meaning, its absolute value has no meaning.
- State function and extensive property
- work, heat: path function

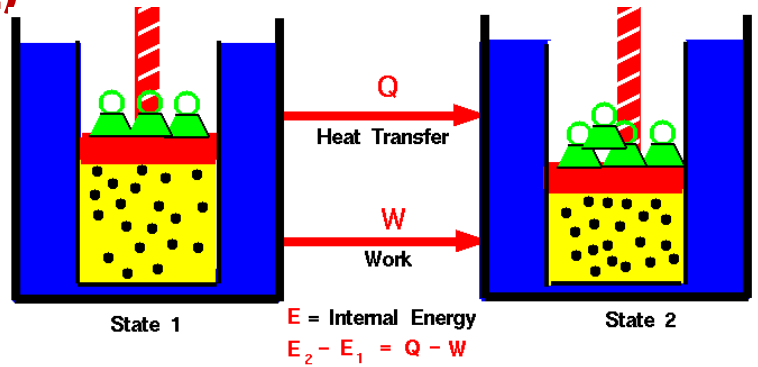
- **Work:** The energy required to move an object against a force.
- **Heat:** The energy transferred from a hot object to a cold object.
- **Path function**: change depends on the path between the initial and final values of the state properties.



The First Law of Thermodynamics

Conservation of Energy (能量守恒)

$$Q - W = \Delta U$$

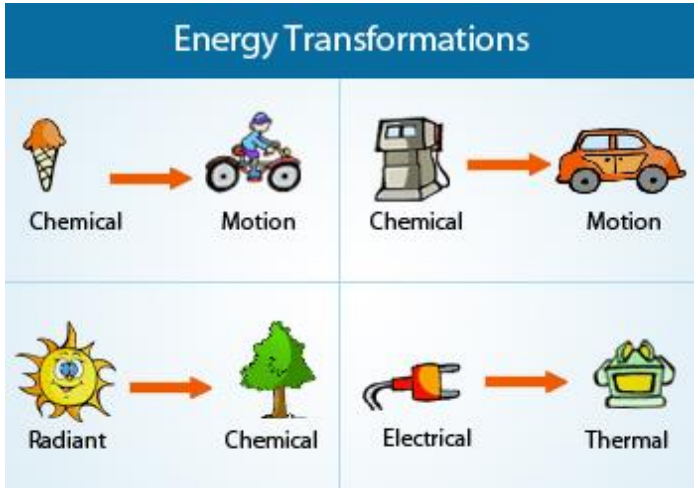


- For a working fluid in a closed system, the heat (*Q*) it receives from the surrounding minus the work (*W*) it performs on the surrounding is equal to the *change* in its *internal energy*.
- $\Delta U = U(2) - U(1)$ $U(2)$: final state; $U(1)$: initial state
- Q > 0: receive heat from the surrounding (*endothermic*，吸热)
- Q < 0: give off heat to the surrounding (*exothermic*，放热)

The First Law of Thermodynamics

Energy Transformation

- The energy may be transformed from one to another.
- The main energy forms are： chemical, mechanical, nuclear, electrical, heat, sound, radiant.
- All energy forms could be set as two type of energy - potential and kinetic energy.
- Internal energy is the total energy of molecules in the working fluid - a sum of kinetic and potential energies.

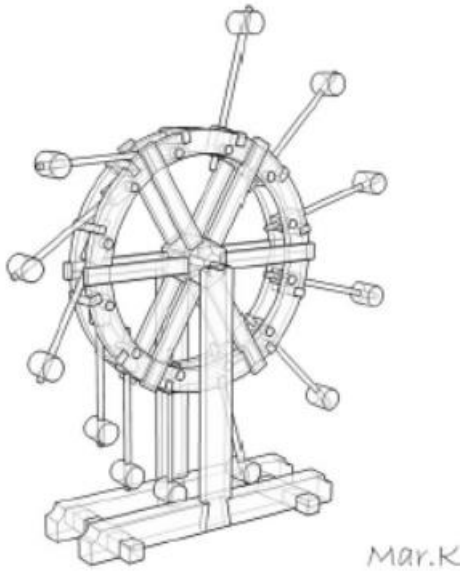




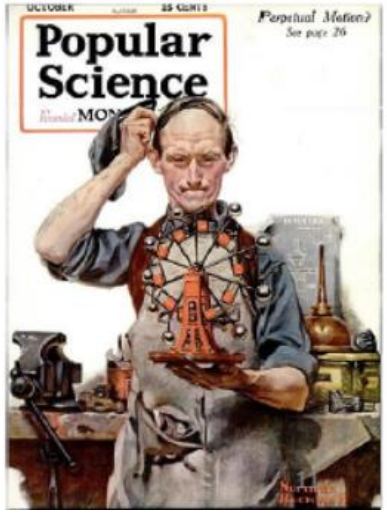
The First Law of Thermodynamics

Perpetuum mobile（永动机）

Perpetuum mobile = perpetual motion machine



- Perpetuum mobile of the first kind (第一类永动机): producing work *indefinitely* with no input of energy.



- First Law ⇒ perpetuum mobile of the first kind does not exist.
- Because internal energy cannot be converted to work in an unlimited way.

Enthalpy (焓)

Enthalpy **is a property of a thermodynamic system. The enthalpy of a system is equal to the system's internal energy plus the product of its pressure and volume.**

The unit of measurement for enthalpy in the International System of Units (SI) is the joule.



Figure 1.15. To create a rabbit out of nothing and place it on the table, the magician must summon up not only the energy U of the rabbit, but also some additional energy, equal to PV, to push the atmosphere out of the way to make room. The *total* energy required is the **enthalpy**, $H = U + PV$.

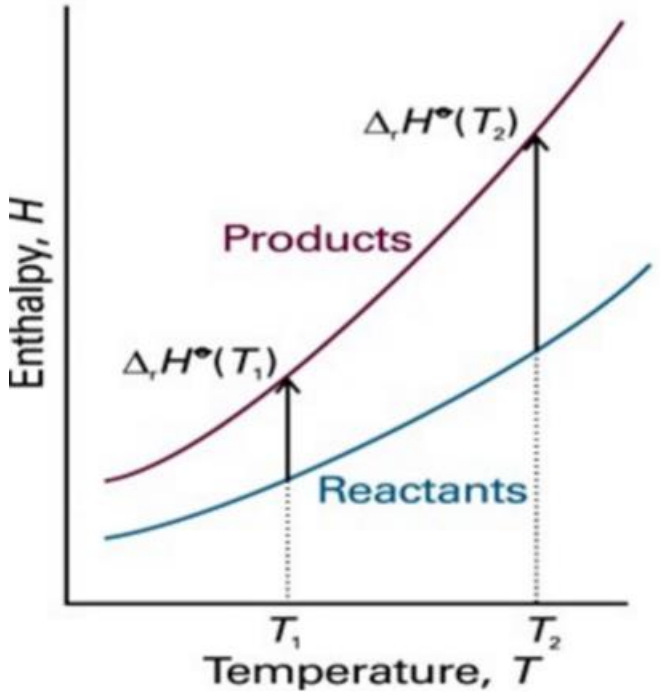
- **Define: *enthalpy* H = *U* + *PV***
- **Enthalpy: a state function that depends only on the prevailing equilibrium state, identified by the variables *internal energy*, *pressure*, and *volume***

Temperature Dependence of Enthalpy

$$\Delta H = H(T_2) - H(T_1) = \int_{T_1}^{T_2} C_p dT$$

$$H(T_2) = H(T_1) + \int_{T_1}^{T_2} C_p dT$$

- If enthalpy is known at temperature T_1, then by integrating C_p the enthalpy can be calculated at T_2.



Standardized Enthalpy & Enthalpy of Formation

In dealing with chemically reacting systems, the concept of standardized enthalpies is extremely valuable. For any species, we can define a *standardized enthalpy* that is the sum of an enthalpy that takes into account the energy associated with chemical bonds, the *enthalpy of formation*, h_f , and an enthalpy that is associated only with the temperature, the *sensible enthalpy change*, Δh_s. Thus, we can write the molar standardized enthalpy for species *i* as

$$\underbrace{\bar{h}_i(T)}_{\substack{\text{Standardized enthalpy}\\ \text{at temperature } T}} = \underbrace{\bar{h}^o_{f,i}(T_{\text{ref}})}_{\substack{\text{Enthalpy of formation}\\ \text{at standard reference}\\ \text{state } (T_{\text{ref}}, P^o)}} + \underbrace{\Delta\bar{h}_{s,i}(T)}_{\substack{\text{Sensible enthalpy}\\ \text{change in going from}\\ T_{\text{ref}} \text{ to } T}},$$

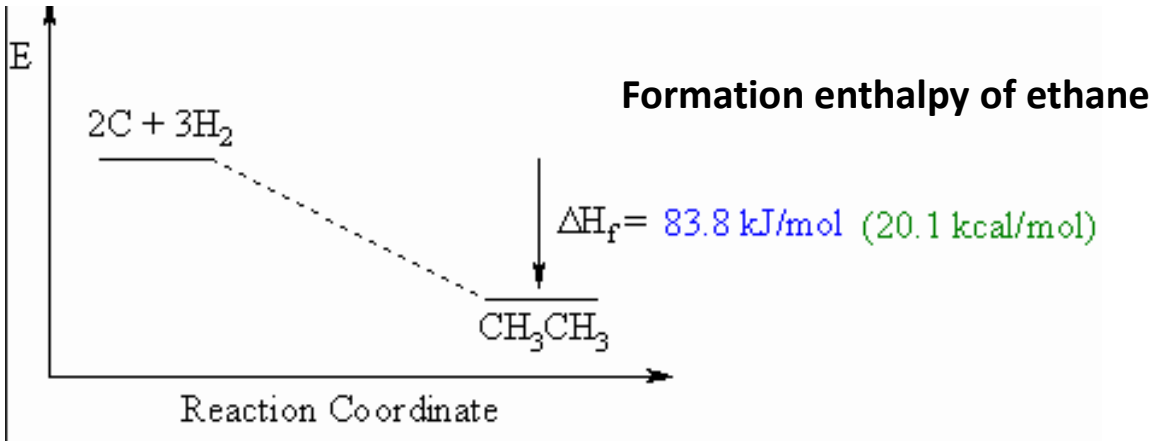
where $\Delta\bar{h}_{s,i} \equiv \bar{h}_i(T) - \bar{h}^o_{f,i}(T_{\text{ref}})$.

Enthalpy of Formation

- The heat released from producing 1 mole of a substance from its elements at a constant temperature T
- The enthalpy of species *k* is such that:

$$h_k = \int_{T0}^{T} C_{p,k}\,dT + \Delta h^0_{f,k}$$

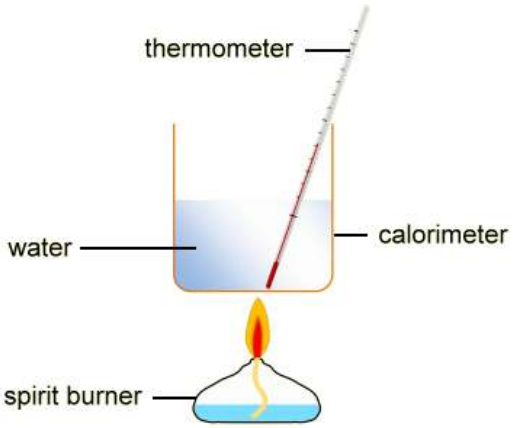
sensible mass enthalpy of formation

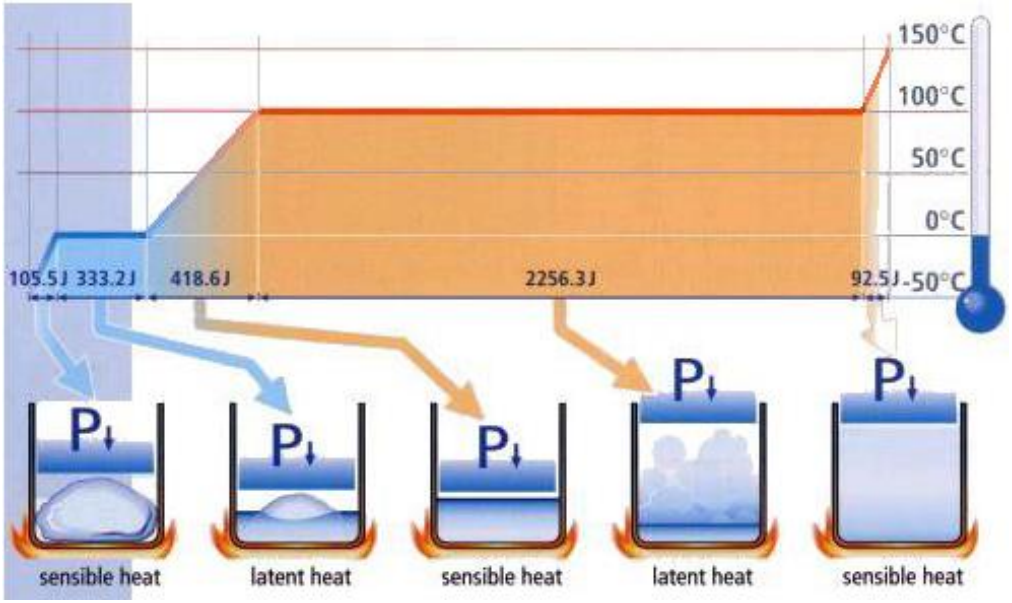


Formation enthalpy of ethane

Three Forms of Enthalpy

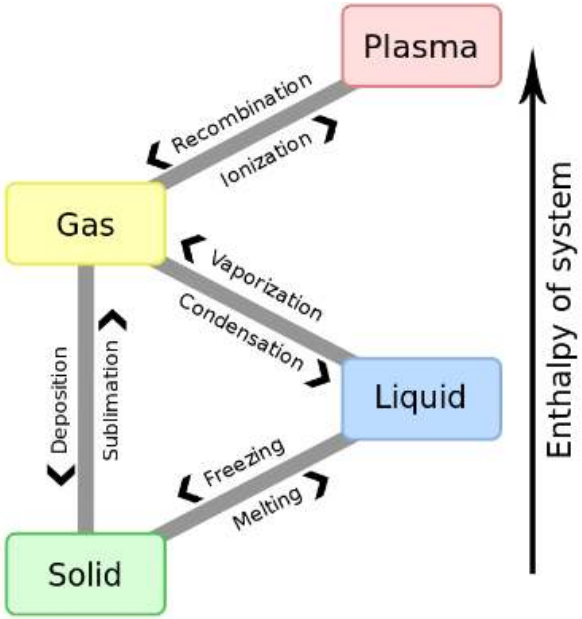
- ***Sensible enthalpy*（显焓）*:*** **When an object is heated, its temperature rises as heat is added. The increase in heat is called sensible heat. Similarly, when heat is removed from an object and its temperature falls, the heat removed is also called sensible heat. Heat that causes a change in temperature in an object is called sensible heat**
- **Examples: heating/cooling**

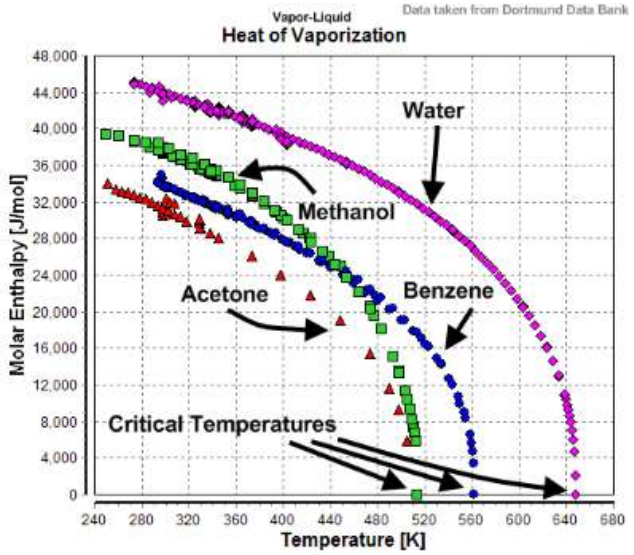




Three Forms of Enthalpy

- ***Latent enthalpy*** **（潜焓）: Latent enthalpy is energy released or absorbed, by a body or a thermodynamic system, during a constant-temperature process.**
- **Examples: phase transition processes: melting, freezing, evaporation/condensation.**

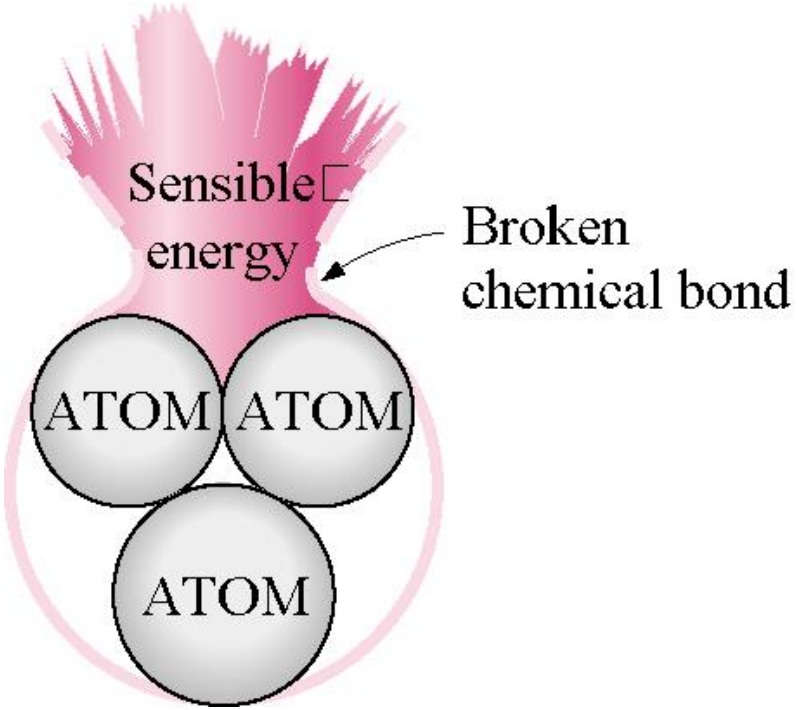


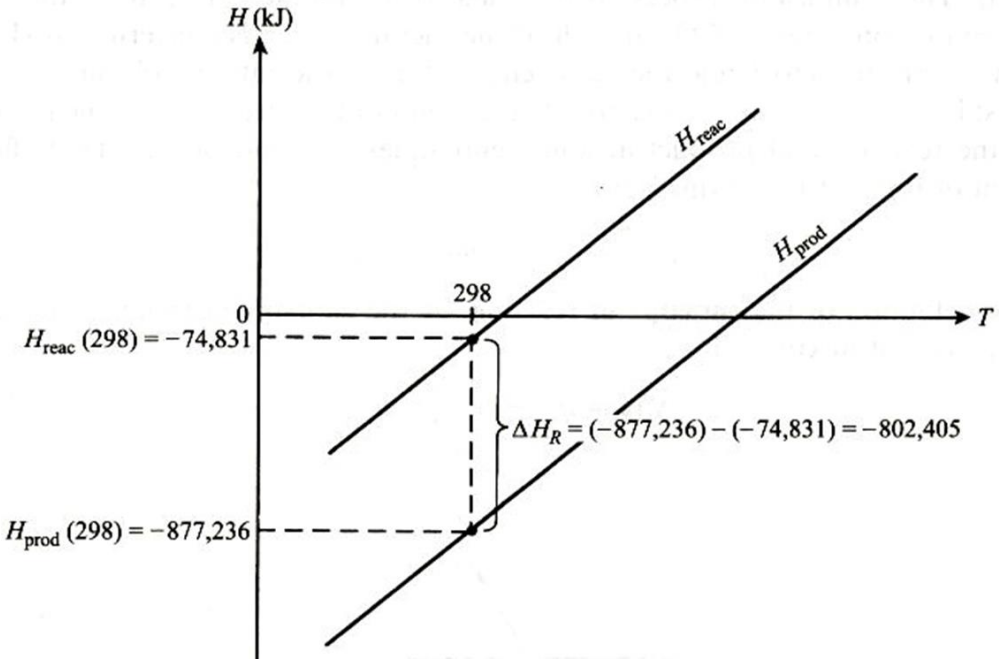


Temperature-dependency of the enthalpy of vaporization for water, methanol, benzene, and acetone.

Three Forms of Enthalpy

- **Reaction enthalpy (反应焓)– reaction enthalpy is the amount of energy or heat absorbed in a reaction.**





Enthalpy of reaction **using representative values for a stoichiometric methane–air mixture. The water in the products is assumed to be in the vapor state.**

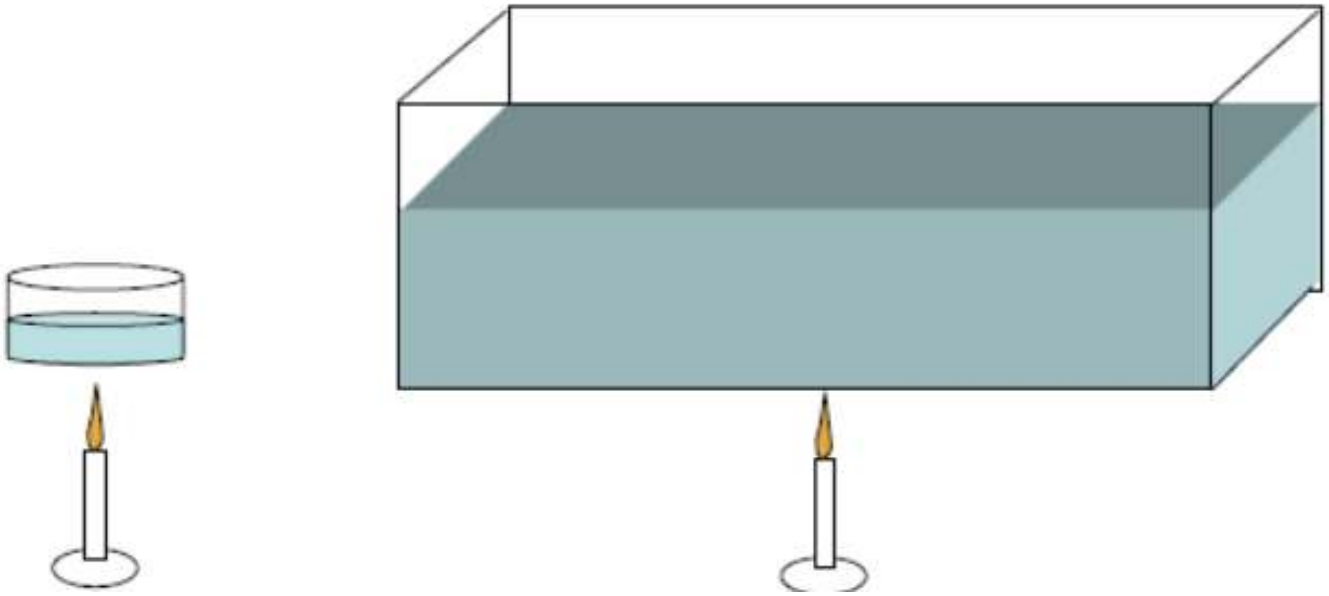
Stand Enthalpy of Formation

Stand enthalpy of formation of some combustion species

Species	W_k (kg/mol)	$\Delta h^o_{f,k}$ (kJ/kg)	$\Delta h^{o,m}_{f,k}$ (kJ/mole)
CH_4	0.016	-4675	-74.8
C_3H_8	0.044	-2360	-103.8
C_8H_{18}	0.114	-1829	-208.5
CO_2	0.044	-8943	-393.5
H_2O	0.018	-13435	-241.8
O_2	0.032	0	0
H_2	0.002	0	0
N_2	0.028	0	0

Table: Molecular mass W_k (kg/mol) and formation enthalpies $\Delta h^o_{f,k}$ (kJ/kg), and $\Delta h^{o,m}_{f,k}$ (kJ/mole) at $T_0 = 298.15$ K.

Heat Capacity



- The *heat capacity* (热容) of a substance is the heat required to change its temperature by one Kelvin, and has units of energy per Kelvin.

Heat Capacity

- For a low-density gas like air or combustion gases, *T*, *P* and *v* are related by the ideal gas law or the equation of state.

$$Pv = RT$$

- The internal energy is primarily a function of *T*.

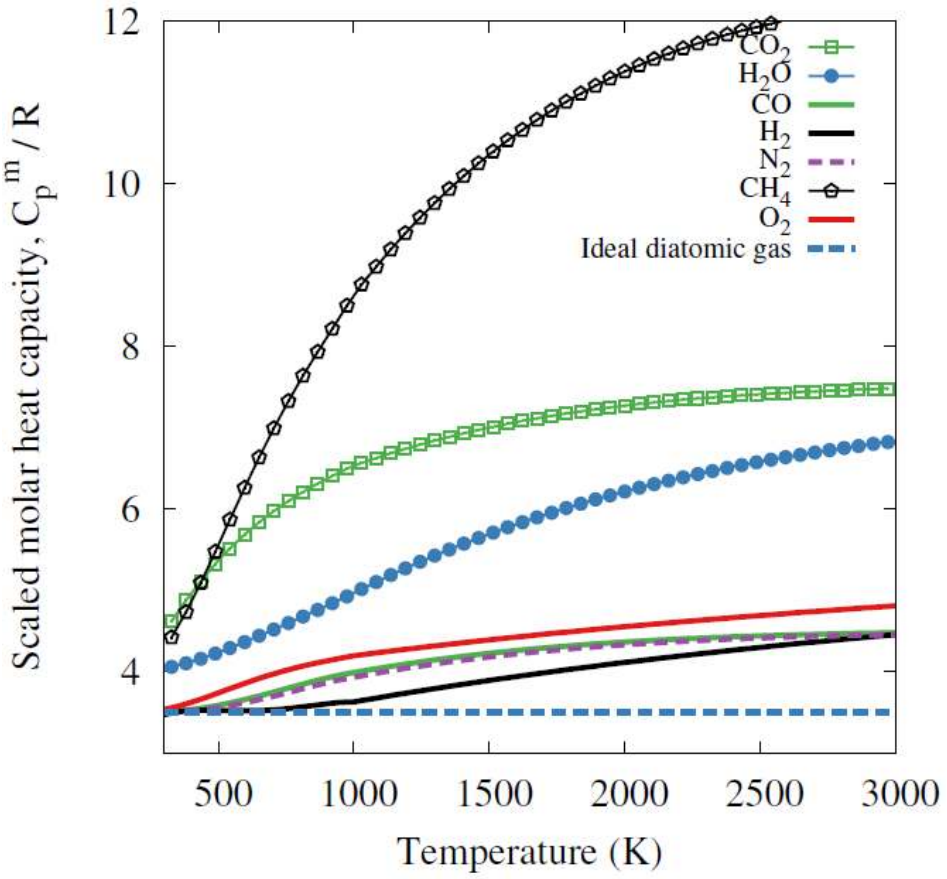
Constant-volume heat capacity $C_v = \left(\frac{\partial u}{\partial T}\right)_v$

- The relationship between enthalpy and temperature may be established by defining a constant-pressure heat capacity.

Constant-pressure heat capacity $C_p = \left(\frac{\partial h}{\partial T}\right)_p$

Relationship of Heat Capacity with T

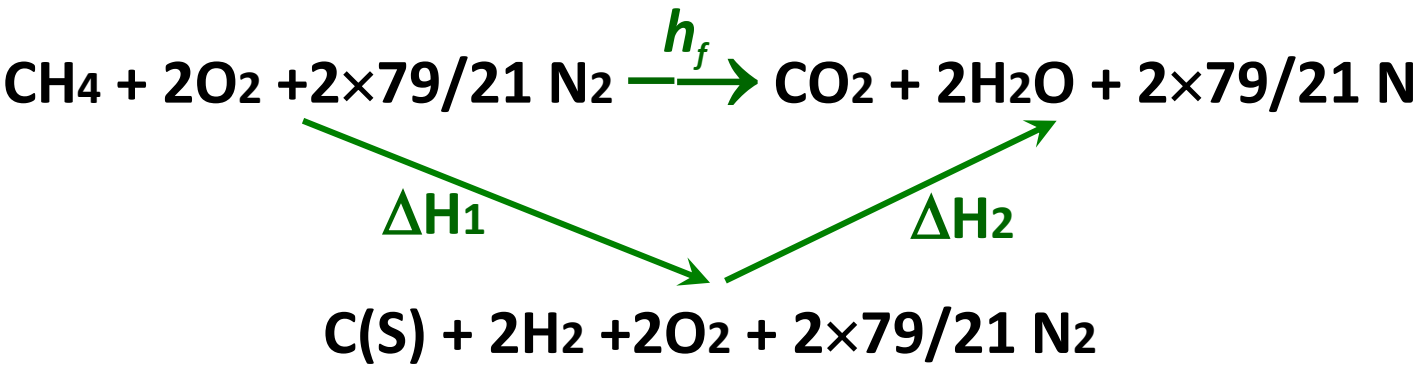
- *Cp* varies with temperature and species.
- The scaled molar heat capacities at constant P ($C_{p,k}^{m}/R$)



- For non-ideal diatomic or polyatomic gaseous molecules, increasing temperature leads to greater population of excited state and higher freedoms of molecules.
- Consequently the $C_{p,k}^{m}$ becomes higher as the temperature increases.

Heat of Combustion

- Definition: ***Heat*** released from complete combustion of 1 mole of a fuel



$$h_f\,(\text{kJ/kmol}) = \Delta H_1 + \Delta H_2 = -1 \times \Delta H^0_{f298}(CH_4) + 1 \times \Delta H^0_{f298}(CO_2) + 2 \times \Delta H^0_{f298}(H_2O)$$

Can also be extended to incomplete combustion system to calculate adiabatic flame temperature ?

Heating Value

- Fuels are classified by their Heating Value, i.e. their heat of combustion
- *Lower Heating Value (LHV)* assumes all the H_2O formed is in gas phase, this is realistic for engines where the H_2O leaves in the exhaust. Note in a real engine the H_2O in the exhaust would be hot, but the LHV calculation usually assumes room temperature steam
- *Upper Heating Value (UHV)* assumes all the H_2O formed is in liquid phase. This is realistic for bomb calorimetry experiments, where the final temperature is usually pretty low.

The Second Law of Thermodynamics

Second-Law Considerations

- ***The First Law of Thermodynamics*** **states that energy cannot be created or destroyed; the total quantity of energy in the universe stays the same.**
- ***The Second Law of Thermodynamics*** **is about the quality of energy. It states that as energy is transferred or transformed, more and more of it is wasted, and there is a natural tendency of any isolated system to degenerate into a more disordered state.**

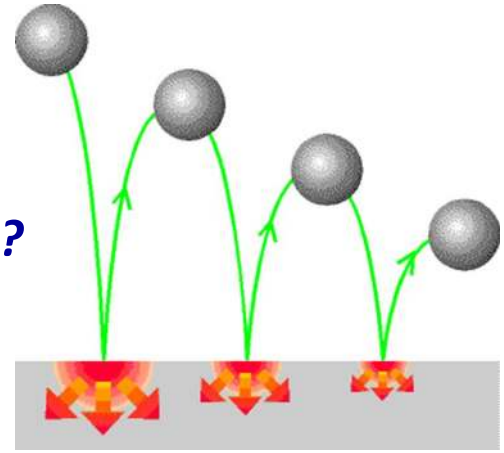
The Second Law of Thermodynamics

Second-Law want to solve:

- ***HOW*** **energy moves around from place to place**
- ***WHY*** **some processes happen and others don't?**
- ***WHAT*****'s the big deal about wanting lowest energy anyway**

The second law provides a guiding principle (law) to predict WHAT things we can do and the maximum extend the things can going on

How do we know that the ball is moving left to right ? why not backwards?



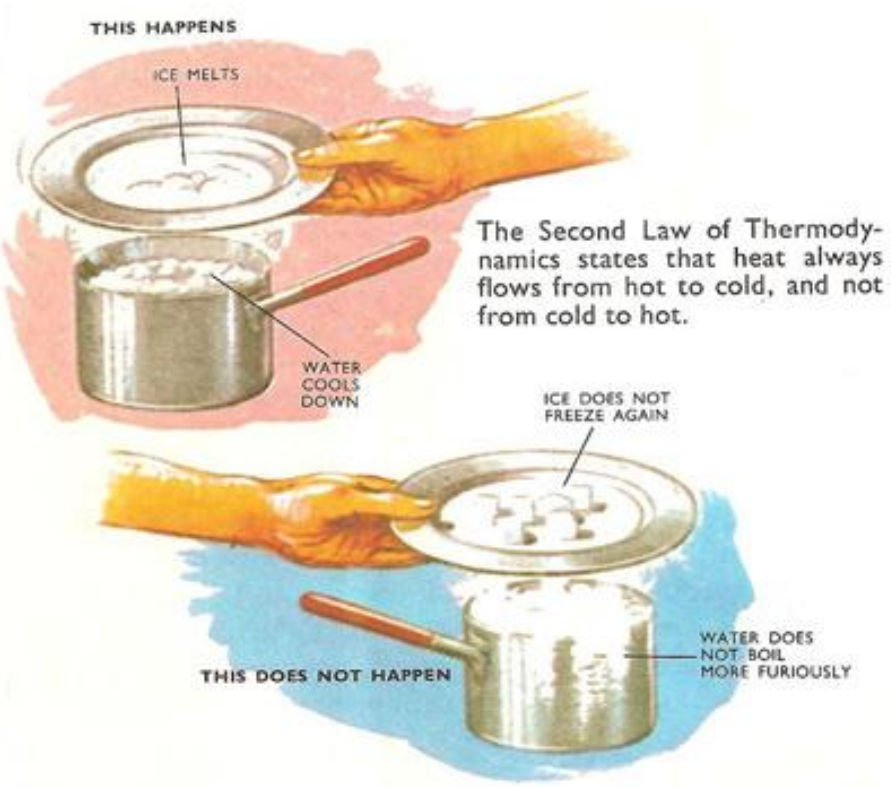
The Second Law of Thermodynamics

The Clausius statement

Heat can never pass from a colder to a warmer body without some other change, connected therewith, occurring at the same time.

The Kelvin statement

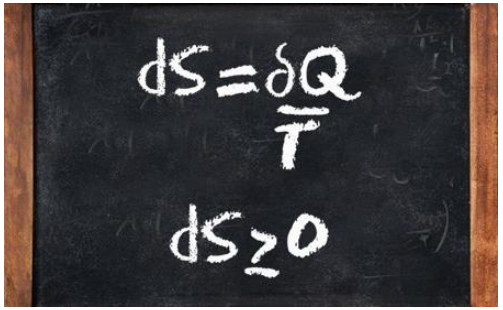
It is impossible, by means of inanimate material agency, to derive mechanical effect from any portion of matter by cooling it below the temperature of the coldest of the surrounding objects.



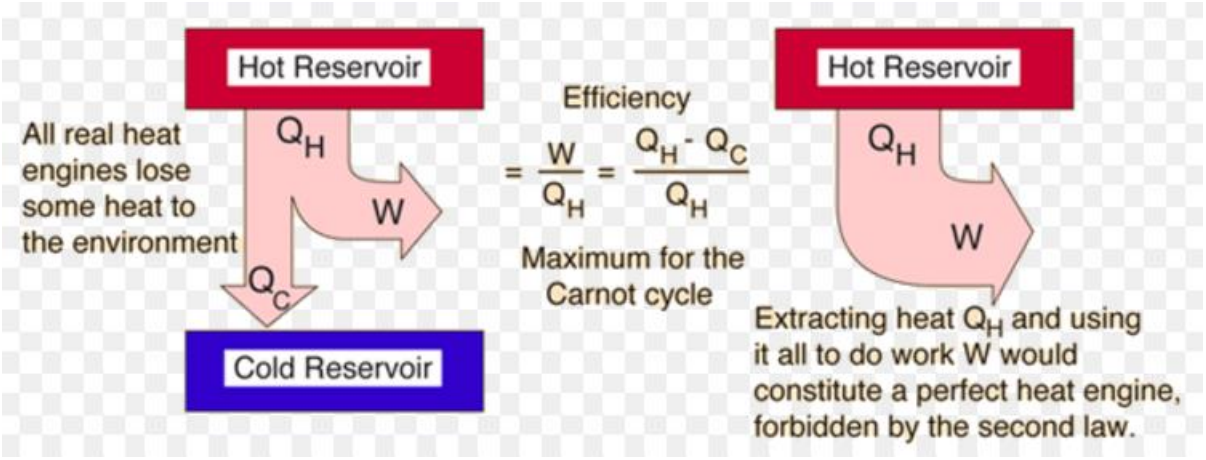
The Second Law of Thermodynamics

- ***The Kelvin-Planck statement***

  It is impossible to construct a device that will operate in a cycle and produce no effect other than the raising of a weight and the exchange of heat with a single reservoir

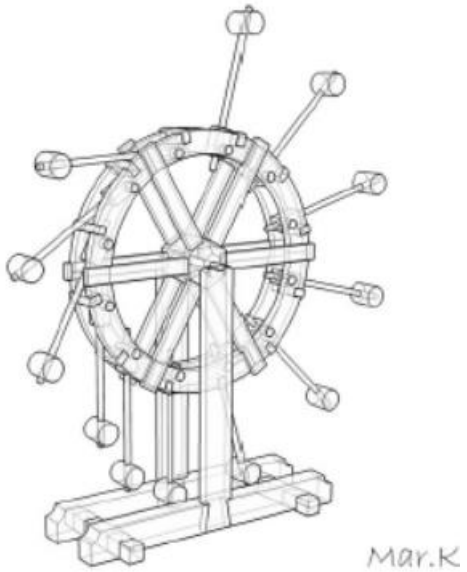


Entropy

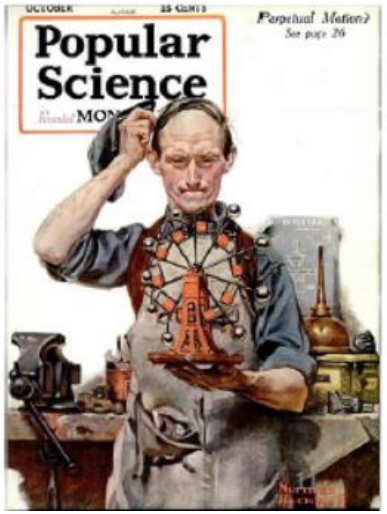


The Second Law of Thermodynamics

Perpetuum mobile?



- Perpetuum mobile of the second kind: 100% conversion of heat to work.
- Second Law ⇒ perpetuum mobile of the second kind does not exist.
- Because it is impossible to construct a device that will operate in a cycle and produce no effect other than the raising of a weight and the exchange of heat with a single reservoir.



Spontaneous Processes

Questions

- ***Why do some processes happen spontaneously?***
- ***Why does a hot body get cooler (rather than hotter) when surrounded by a cooler medium?***
- ***Why does a gas expand into all available volume of a container rather than contract?***

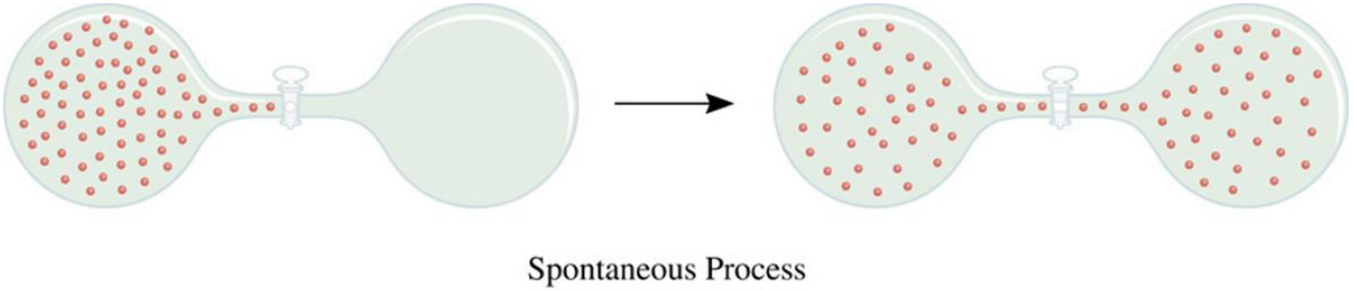
- **The driving force for spontaneous change (change that happens without intervention of doing work or heating) is described in the second law of thermodynamics.**

Entropy (熵)

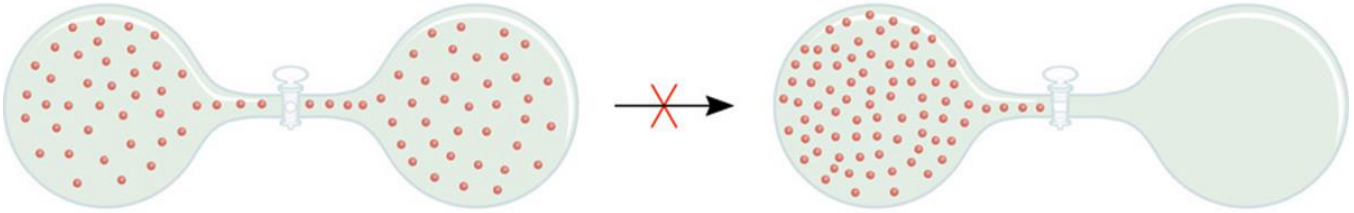
- From the point of statistical thermodynamics: ***entropy*** is a measure of molecular randomness.

$$S = \left(\frac{\delta Q}{T}\right)_{int\ rev}$$

$(\delta Q)_{int\ rev}$: the heat a control mass received during an infinitesimal, internally reversible process



Spontaneous Process



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Nonspontaneous Process

Entropy

- For a spontaneous process to occur, the entropy of the control mass must be equal to or greater than zero, that is, for a spontaneous process to occur, the entropy of the control volume must increase.

$$\Delta S = S_2 - S_1 \geq 0$$

- Without heat release or absorption, a spontaneous process will proceed in the direction to increase entropy.

The Second Law and Entropy

Entropy — S (T, P)

- **Apply the first law to a reversible constant T, P process**

$$\delta Q_{int\,rev} - \delta W_{int\,rev} = dU$$

$$\delta Q_{int\,rev} = TdS$$

$$\delta W_{int\,rev} = PdV$$

$$TdS = dU + PdV \qquad dS = \frac{du}{T} + \frac{Pdv}{T}$$

$$ds = C_v\frac{dT}{T} + R\frac{dv}{v} \qquad ds = C_p\frac{dT}{T} + R\frac{dp}{p}$$

S increases as *T* increases, but decreases with *P* increases

The third law: S = 0 at 0 K

$$\Delta s = s_2 - s_1 = \int_1^2 \left(C_v\frac{dT}{T}\right) + R\ln\frac{v2}{v1} \qquad \Delta s = s_2 - s_1 = \int_1^2 \left(C_p\frac{dT}{T}\right) + R\ln\frac{p2}{p1}$$

- ***Standard-state entropy (1 atm)*:** $\Delta s = \int_0^T \left(C_p\frac{dT}{T}\right)$

Chemical Equilibrium

$CH_4 + 2O_2 + (2\times79/21)\,N_2 \rightarrow CO_2 + 2H_2O + (2\times79/21)\,N_2$

Complete combustion, greater heat release

$CH_4 + 2O_2 + (2\times79/21)\,N_2 \rightarrow CO_2 + 2H_2 + O_2 + (2\times79/21)\,N_2$

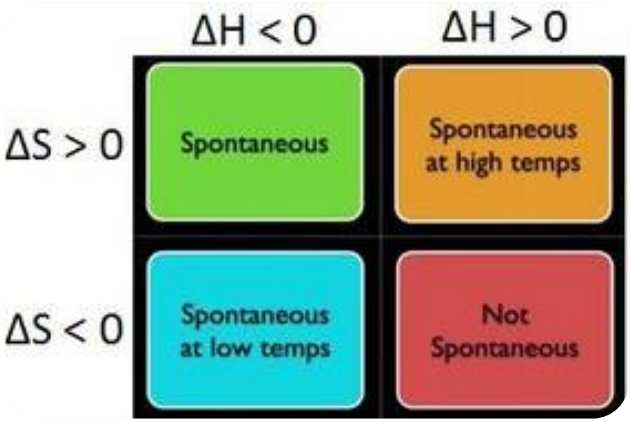
Incomplete combustion, less heat release, more entropy

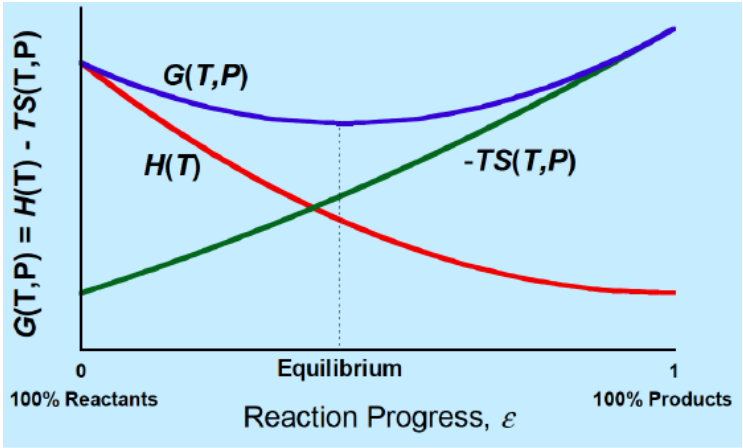
- In practical combustion processes, a combustion reaction can never reach completion. Rather the products of combustion will acquire the state of ***chemical equilibrium.***
- A compromise between enthalpy and entropy releases must be made!
- ***Q: How does the reaction know when to stop?***

Gibbs Free Energy

Gibbs free energy is a thermodynamic potential that can be used to calculate the maximum of reversible work that may be performed by a thermodynamic system at a constant T and P. The Gibbs free energy (G = H-TS) is the *maximum* amount of non-expansion work that can be extracted from a thermodynamically closed system; this maximum can be attained only in a completely reversible process.

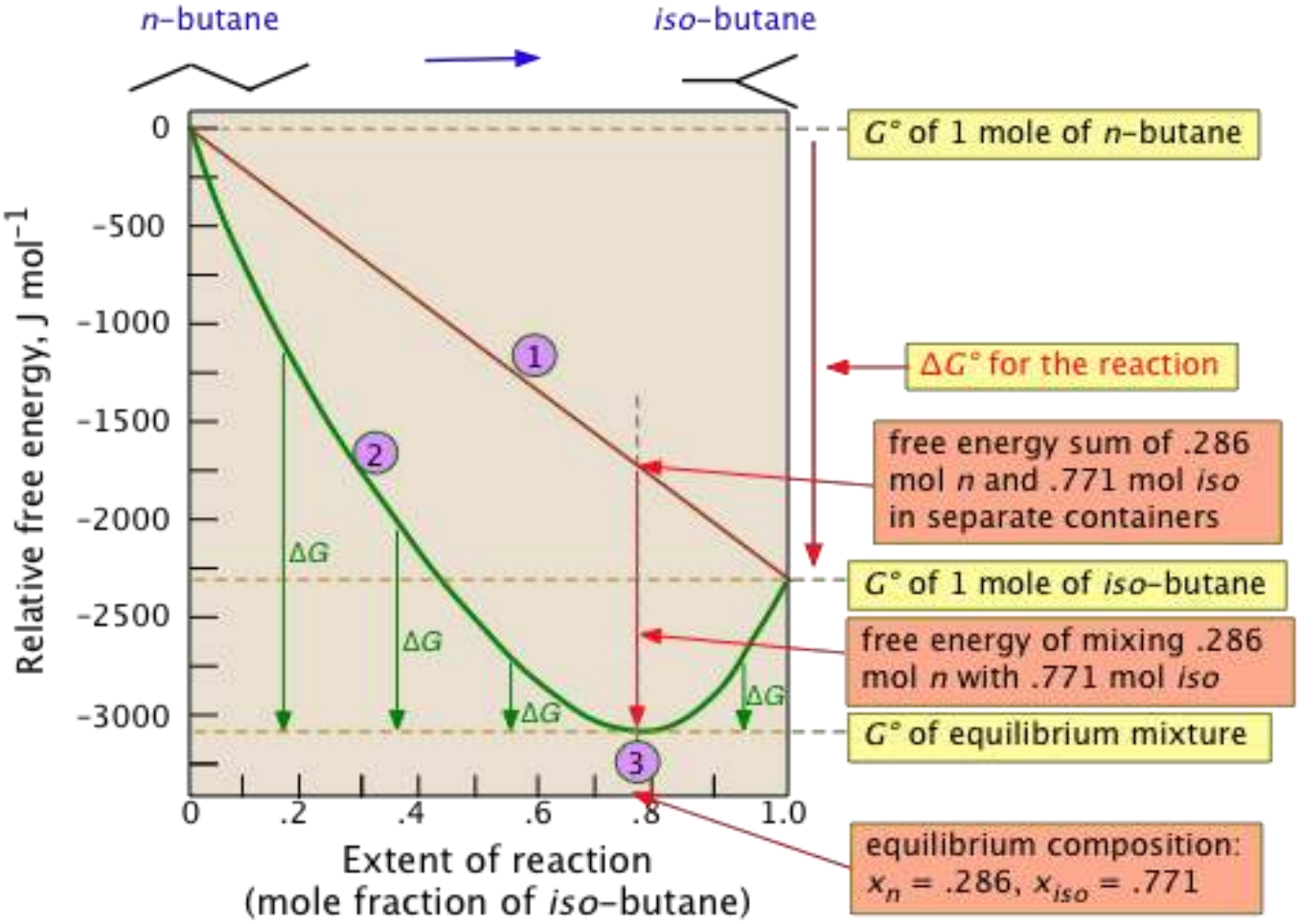
$$G = H - TS$$
$$\Delta G = \Delta H - T\Delta S$$





- The free enthalpy G (*Gibbs energy*) indicates if a reaction is spontaneously possible or not
- Equilibrium is found when $dG/d\varepsilon = 0$

Gibbs Free Energy - Example



- The equilibrium composition corresponding to ~77 % conversion
- G determines the extend of a reaction processes

Equilibrium Constant

Reaction $\sum_{react.} Vi\,Ai \rightarrow \sum_{prod.} Vi\,Ai$

Gibbs free energy

$$G(T,P) = \sum_{react.} ni\,gi\ (\mathrm{T,P}) + \sum_{prod.} ni'\,gi'\ (\mathrm{T,P})$$

Equilibrium at const. T, P

$$\frac{dG(T,P)}{d\varepsilon} = 0$$

$$= \sum_{react.} g_i\ (\mathrm{T,P})\frac{dm_i}{d\varepsilon} + \sum_{prod.} g_{i'}\ (\mathrm{T,P})\frac{dm_{i'}}{d\varepsilon}$$

Mass conversion

$$-\sum_{react.} v_i\,g_i\ (\mathrm{T,P}) + \sum_{prod.} v_{i'}g_{i'}\ (\mathrm{T,P}) = 0$$

Gibbs function of species *i*

$$g_i(T,P) = g^{0(T)} + RT\ln\left(\frac{P}{P_0}\right)$$

.....

$$\Delta G_r^0(T) = -R_u T \ln\left(\frac{\prod_{prod.} P_{i'}^{\nu_{i'}}}{\prod_{react.} P_i^{\nu_i}}\right)$$

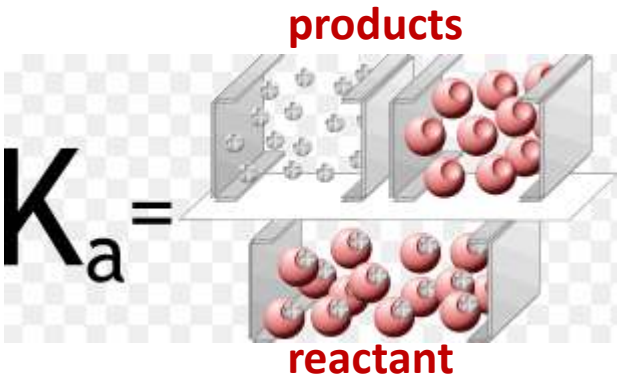
$$K_p(T) \equiv \frac{\prod_{prod.} P_{i'}^{\nu_{i'}}}{\prod_{react.} P_i^{\nu_i}} = \exp\left[-\frac{\Delta G_r^0(T)}{R_u T}\right]$$

Equilibrium Constant

Equilibrium Constant

- Kp is a function of temperature only

$$K_p(T) \equiv \frac{\prod_{\text{prod.}} P_i^{\nu_i}}{\prod_{\text{react.}} P_i^{\nu_i}} = \exp\left[-\frac{\Delta G_r^{\circ}(T)}{R_u T}\right]$$

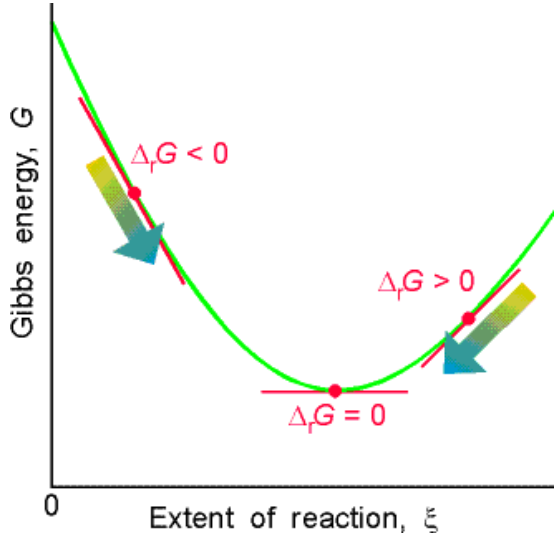


- Forward/backward equilibrium constant

$$H_2O = H_2 + \tfrac{1}{2}\,O_2 \qquad H_2 + \tfrac{1}{2}\,O_2 = H_2O$$

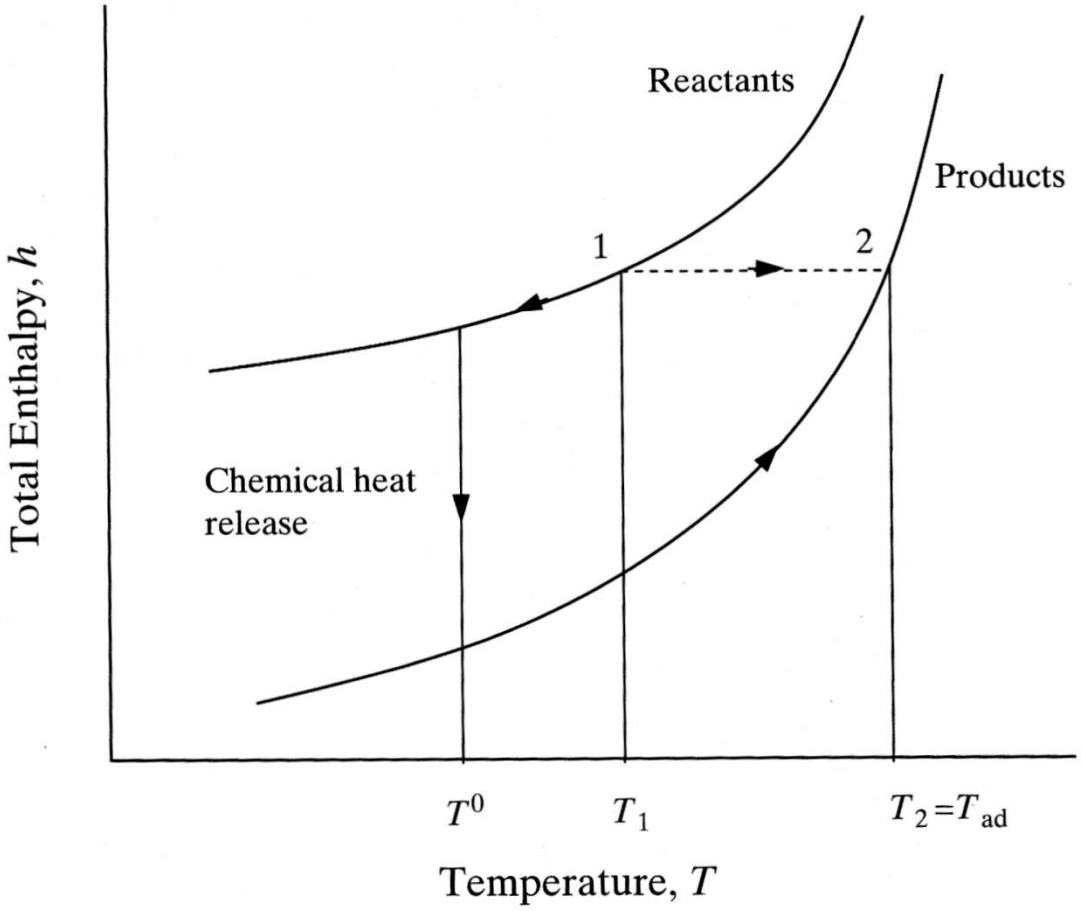
$$K_{p,f}(T) = \frac{P_{H_2} P_{O_2}^{1/2}}{P_{H_2O}} \qquad K_{p,b}(T) = \frac{P_{H_2O}}{P_{H_2} P_{O_2}^{1/2}}$$

$$K_{p,f}(T) = \frac{1}{K_{p,b}(T)}$$

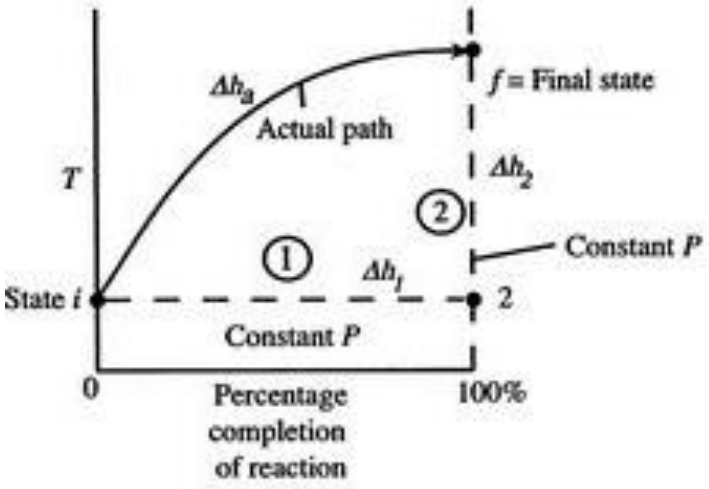


- The larger the equilibrium constant, the greater the completion of a reaction

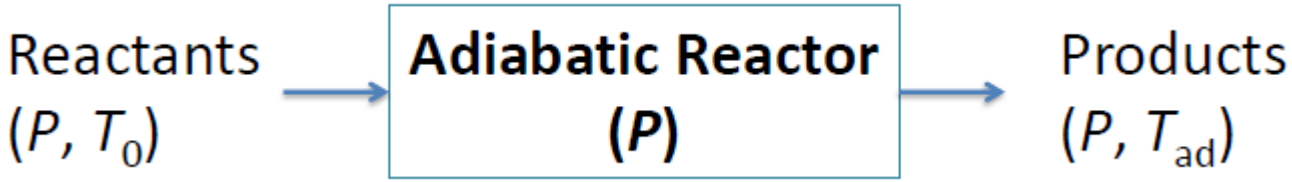
Adiabatic Flame Temperature



- The ***adiabatic flame temperature*** **is obtained when all the heat released from combustion reaction is used to raise the product temperature from T_0 to T_{ad}.**



Adiabatic Flame Temperature



- **Adiabaticity gives** $$H_{\text{prod.}}\left(T_{ad}\right) - H_{\text{react.}}\left(T_0\right) = 0$$

- **Using the total enthalpy equation for each species**

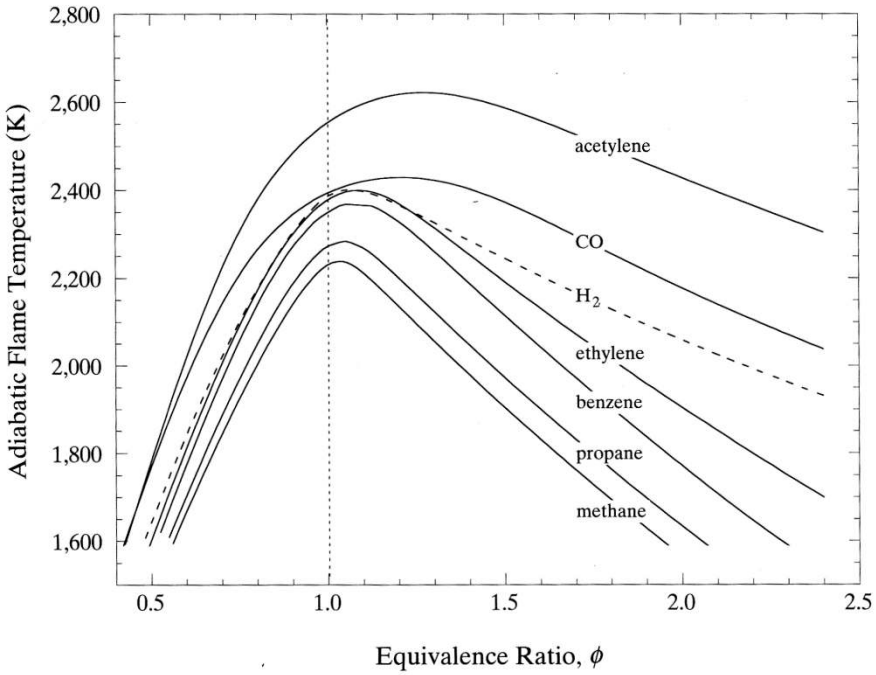
$$\sum_{\text{prod.}} \nu_{i'} \bar{h}_{T_{ad},i'} - \sum_{\text{react.}} \nu_i \bar{h}_{T_0,i} = \left[\sum_{\text{prod.}} \nu_{i'} \bar{h}^{\circ}_{f,298,i'} - \sum_{\text{react.}} \nu_i \bar{h}^{\circ}_{f,298,i} \right] + \sum_{\text{prod.}} \nu_{i'} \left[\bar{h}\left(T_{ad}\right) - \bar{h}\left(298\right) \right]_{i'}$$

$$- \sum_{\text{react.}} \nu_i \left[\bar{h}\left(T_0\right) - \bar{h}\left(298\right) \right]_i = 0$$

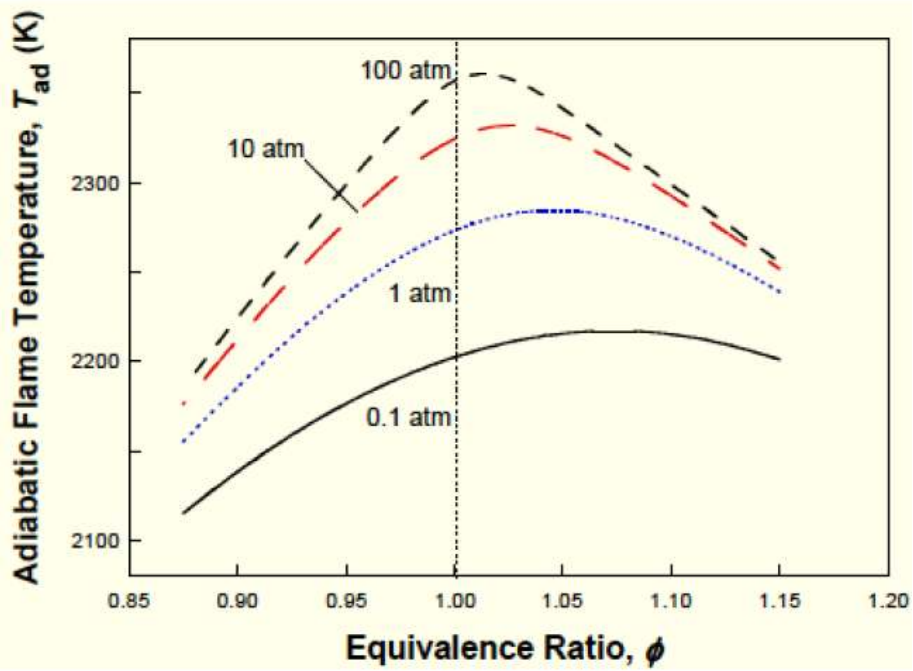
- **Let T_0 = 298 K** $$-\Delta H^{\circ}_{r,298} = \sum_{\text{prod.}} \nu_{i'} \left[\bar{h}\left(T_{ad}\right) - \bar{h}\left(298\right) \right]_{i'}$$

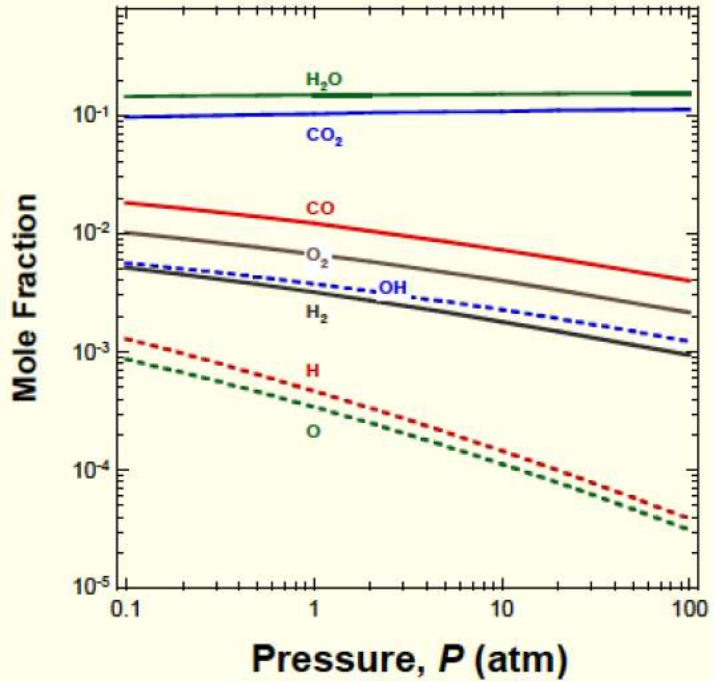
Adiabatic Flame Temperature

- The decrease of T_{ad} towards smaller ϕ (equivalence ratio) is caused by dilution of unused oxygen as well as the greater amount of nitrogen brought into the combustor with oxygen.
- The decrease of the flame temperature towards larger ϕ is because of oxygen deficiency, which does not allow the fuel to be fully oxidized.



Adiabatic Flame Temperature

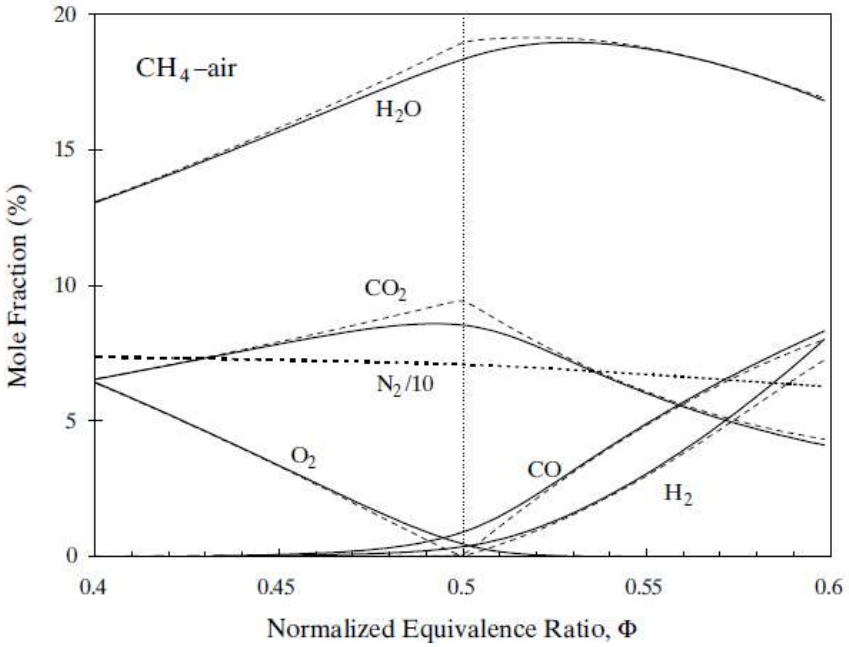




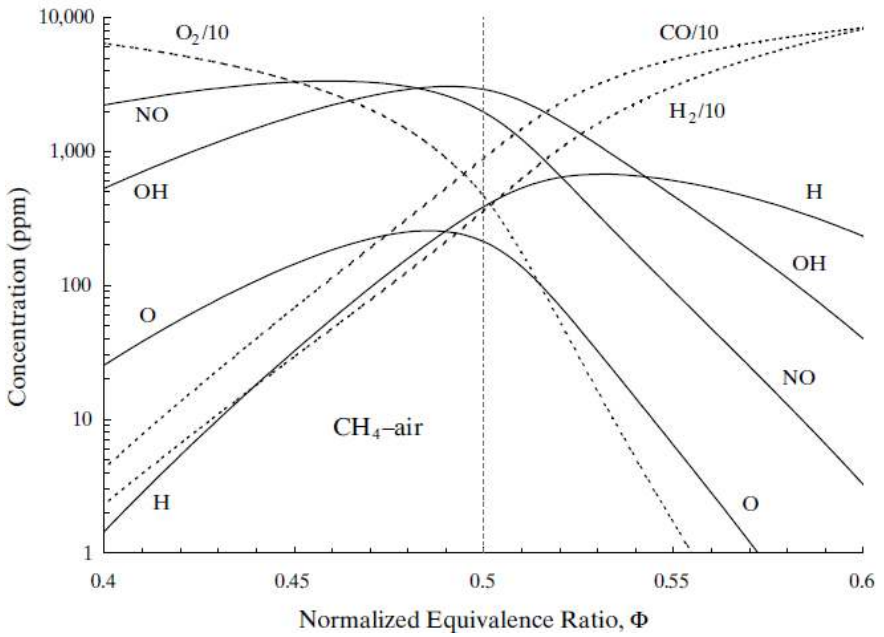
- As pressure increases, (a) the adiabatic flame temperature becomes higher and (b) the peak shifts towards $\varphi = 1$.
- The pressure serves to reduce the extent of dissociation of CO_2 and H_2O, and forces the reaction towards better completion.

Equilibrium Composition – CH_4/air Flame

Major flame species



Minor flame species



reaction DESIGN CHEMKIN

Gaseq

www.gaseq.co.uk

Chemical Equilibrium with Applications

www.grc.nasa.gov/WWW/CEAWeb

A lot of commercial or online calculation programs

Summary of Thermodynamics Equations

Process	Property	Process formula	Energy relationship	Internal energy ΔE	work A	heat Q	Heat capability C
Constant V	V = c	P/T=c	$Q=\Delta E$	$\frac{M}{\mu}C_V(T_2-T_1)$	**0**	$\frac{M}{\mu}C_V(T_2-T_1)$	$C_V=\frac{i}{2}R$
Constant P	P = c	V/T=c	$Q=\Delta E+A$	$\frac{M}{\mu}C_V(T_2-T_1)$	$P(V_2-V_1)$ **or** $\frac{M}{\mu}R(T_2-T_1)$	$\frac{M}{\mu}C_P(T_2-T_1)$	$C_P=C_V+R$
Constant T	T = c	PV=c	$Q=A$	**0**	$\frac{M}{\mu}RT\ln\frac{V_2}{V_1}$ **or** $\frac{M}{\mu}RT\ln\frac{P_1}{P_2}$	$\frac{M}{\mu}RT\ln\frac{V_2}{V_1}$ **or** $\frac{M}{\mu}RT\ln\frac{P_1}{P_2}$	∞
Adiabat	Q = 0	$PV^{\gamma}=C_1$ $V^{\gamma-1}T=C_2$ $P^{\gamma-1}V^{-\gamma}=C_3$	$A=-\Delta E$	$\frac{M}{\mu}C_V(T_2-T_1)$	$-\frac{M}{\mu}C_V(T_2-T_1)$	**0**	**0**

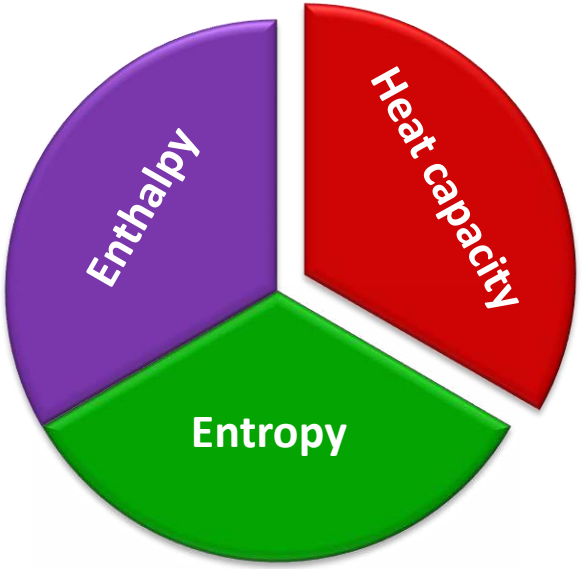
Outline of Chapter 2

- Introduction to Thermodynamics
- Fundamentals of Thermodynamics
- Thermodynamics in Combustion
- Transport Properties
- Transport in Combustion

Thermodynamics Data

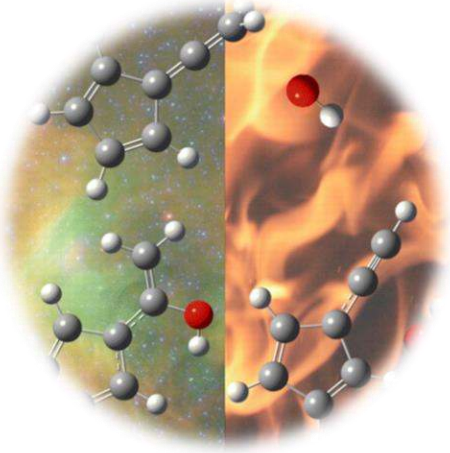
Parameters needed in kinetic modeling

- Thermochemical data for all species
  - Enthalpy of formation ΔH_f^0
  - Entropy S^0
  - Heat capacity C_p^0



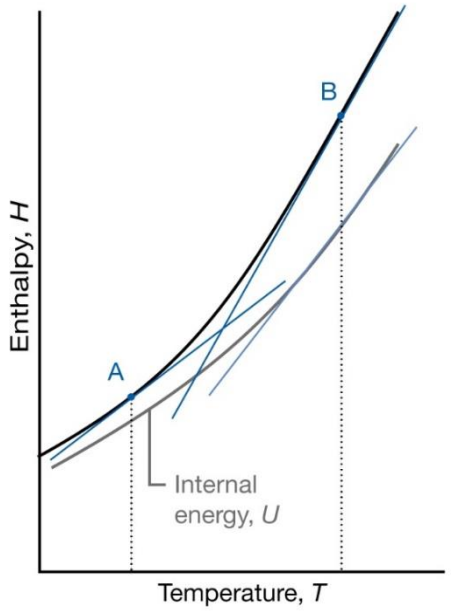
Necessary for the calculation of

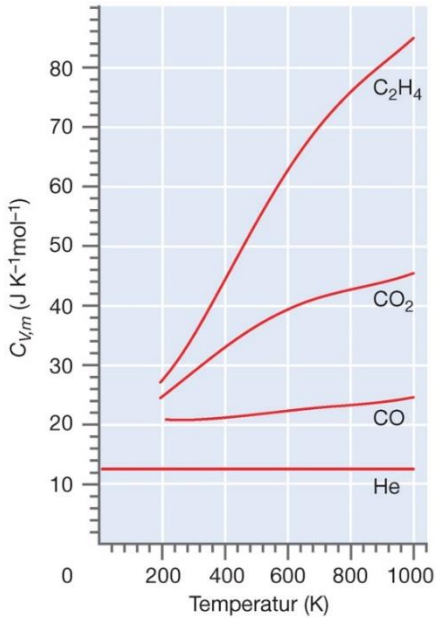
- Thermal effect of the reaction
- Energy conversion equation
- Equilibrium constant of the reaction
- Equilibrium composition and T_{ad}
- Relationship between direct and forward rate constant of an elementary reaction



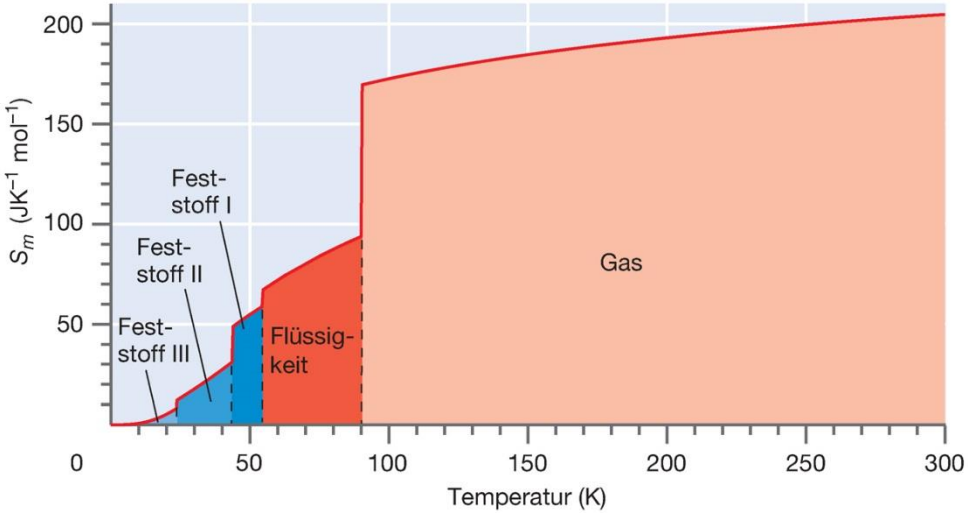
Temperature Dependence

Enthalpy



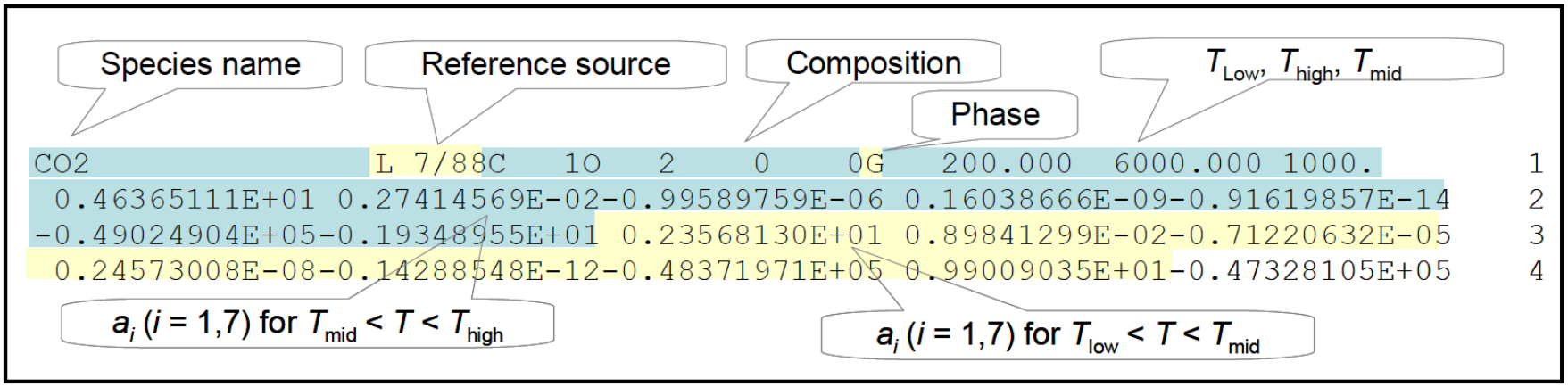


Heat capacity



Entropy

NASA Polynomials



$$\overline{c}_p/R_u = a_1 + a_2T + a_3T^2 + a_4T^3 + a_5T^4,$$

$$\overline{s}^\circ/R_u = a_1 \ln T + a_2T + a_3T^2/2 + a_4T^3/3 + a_5T^4/4 + a_7.$$

$$\overline{h}_T/R_u = a_1T + a_2T^2/2 + a_3T^3/3 + a_4T^4/4 + a_5T^5/5 + a_6$$

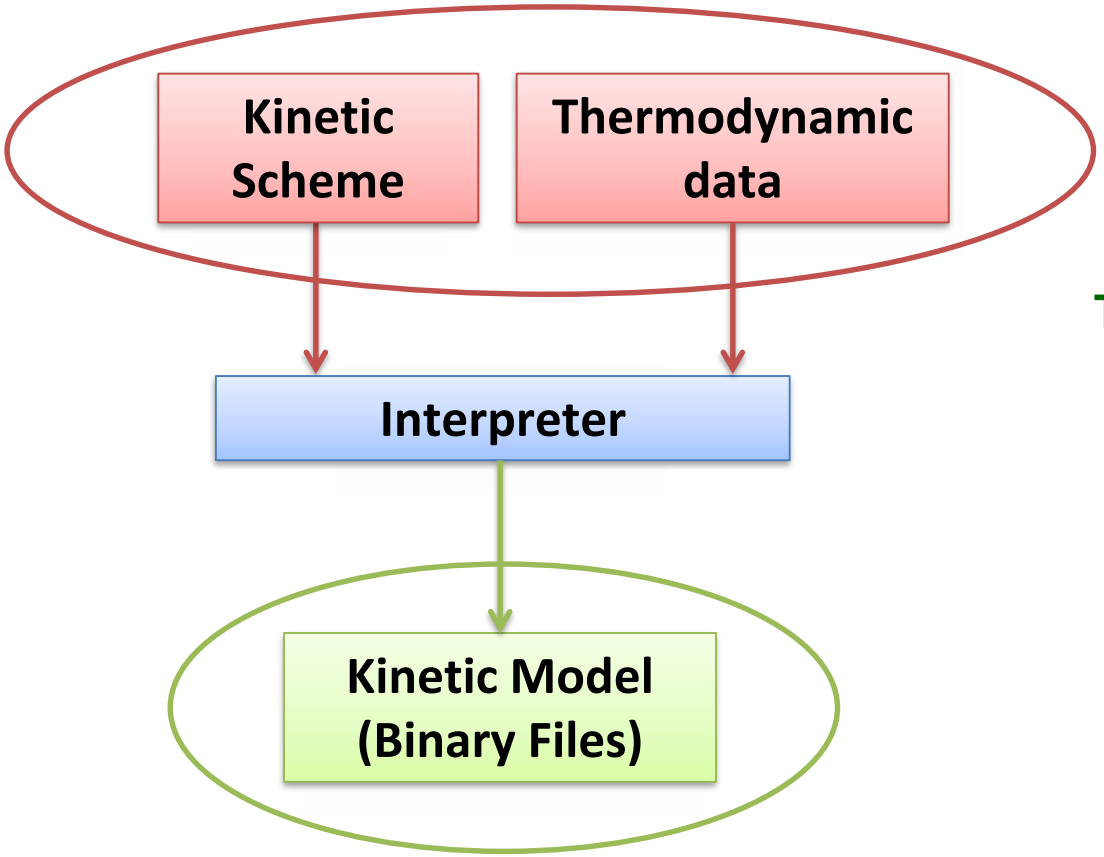
$$\overline{h}(T) - \overline{h}(298.15) = \overline{h}_T(T) - \overline{h}_T(298.15)$$

$$\overline{h}_f^\circ(T) = \overline{h}_T(T) - \sum_{\text{elements}} \nu_i \overline{h}_{T,i}(T)$$

- Adopted in many databases
- Used in almost all computational codes

Interpretation of Thermodynamic Data

CHEMKIN/CANTERA/OPENSMOKE/FLAMEMASTER/COSILAB/LOGE/FLUENT……



Readable for codes

Thermodynamic properties

- Species enthalpy
- Species entropy
- Species C_p
- Formation enthalpy
- Gibbs energy
- Equilibrium constant
- ……

Source of Thermodynamic Data

Online databases

- http://webbook.nist.gov/chemistry/
- http://kinetics.nist.gov/janaf/
- http://atct.anl.gov/
- http://garfield.chem.elte.hu/burcat/burcat.html

NIST-JANAF Thermochemical Tables

Methane (CH_4) $C_1H_4(g)$

Enthalpy Reference Temperature = T_r = 298.15 K Standard State Pressure = p° = 0.1 MPa

	$J \cdot K^{-1}\ mol^{-1}$			$kJ \cdot mol^{-1}$			
T/K	C_p°	S°	$-[G^\circ - H^\circ(T_r)]/T$	$H - H^\circ(T_r)$	$\Delta_f H^\circ$	$\Delta_f G^\circ$	log K_f
0	0.	0.	INFINITE	-10.024	-66.911	-66.911	INFINITE
100	33.258	149.500	216.485	-6.698	-69.644	-64.353	33.615
200	33.473	172.577	189.418	-3.368	-72.027	-58.161	15.190
250	34.216	180.113	186.829	-1.679	-73.426	-54.536	11.395
298.15	35.639	186.251	186.251	0.	-74.873	-50.768	8.894
300	35.708	186.472	186.252	0.066	-74.929	-50.618	8.813
350	37.874	192.131	186.694	1.903	-76.461	-46.445	6.932
400	40.500	197.356	187.704	3.861	-77.969	-42.054	5.492
450	43.374	202.291	189.053	5.957	-79.422	-37.476	4.350
500	46.342	207.014	190.614	8.200	-80.802	-32.741	3.420
600	52.227	215.987	194.103	13.130	-83.308	-22.887	1.993
700	57.794	224.461	197.840	18.635	-85.452	-12.643	0.943
800	62.932	232.518	201.675	24.675	-87.238	-2.115	0.138
900	67.601	240.205	205.532	31.205	-88.692	8.616	-0.500
1000	71.795	247.549	209.370	38.179	-89.849	19.492	-1.018

Source of Thermodynamic Data

Research papers

AIP | Journal of Physical and Chemical Reference Data

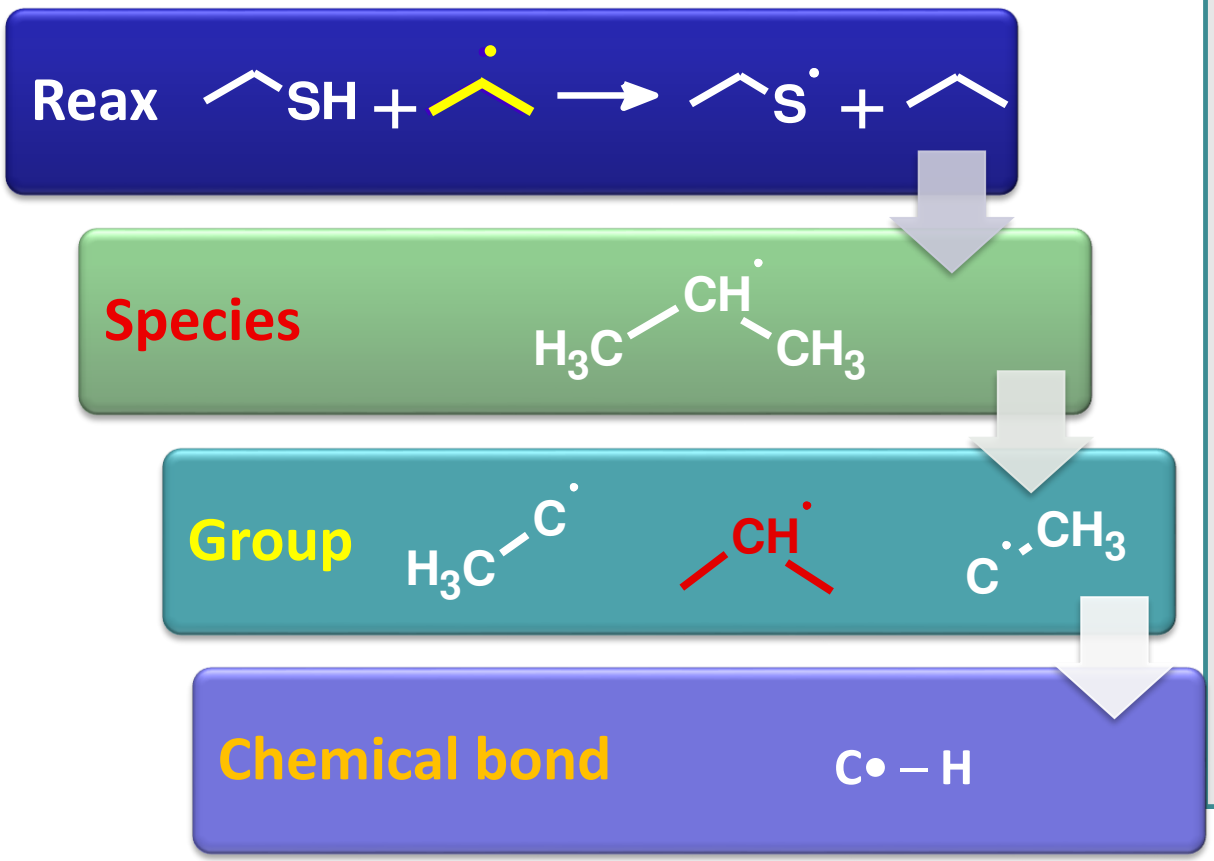
Critical Evaluation of Thermochemical Properties of C1–C4 Species: Updated Group-Contributions to Estimate Thermochemical Properties

S. M. Burke, J. M. Simmie, and H. J. Curran

Citation: Journal of Physical and Chemical Reference Data **44**, 013101 (2015); doi: 10.1063/1.4902535
View online: http://dx.doi.org/10.1063/1.4902535
View Table of Contents: http://scitation.aip.org/content/aip/journal/jpcrd/44/1?ver=pdfcov
Published by the AIP Publishing

Source of Thermodynamic Data

Group Additivity Method



$$K_r = e^{-\frac{\Delta_r G^\circ}{RT}}$$

$$\Delta_r G^\circ = \sum_i v_i \, \Delta_f G^\circ(i)$$

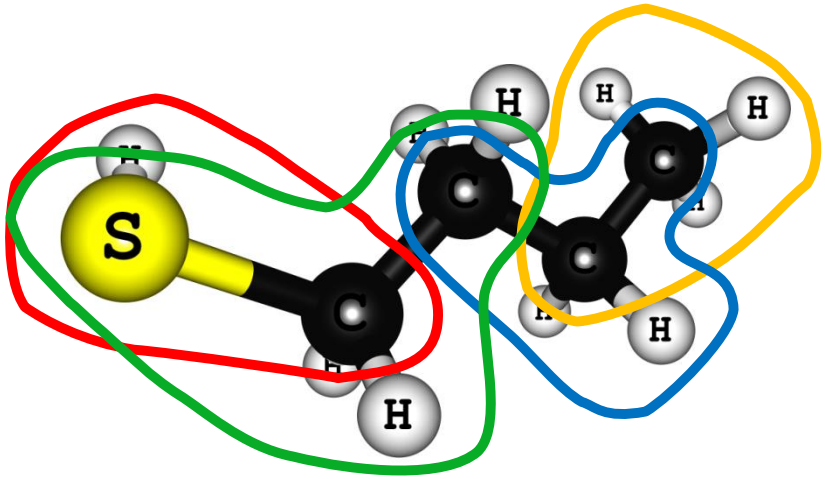
$$\Delta_f G^\circ(i) = \sum_g \Delta_{gr} G^\circ(g)$$

$$\Delta_{gr} G^\circ(g) = \sum_b \Delta_{bond} G^\circ(b)$$

- The thermochemical properties of a species are additive with respect to the functional groups that makes up the molecule

Source of Thermodynamic Data

Benson's Group Additivity Method



- Cutting of molecules into groups

Groups:

1) Atom with a valence ≥ 2;
2) Atoms or groups of monovalent atoms bonded to it.

ex.: C-(C)(H)3
C-$(C)_2(H)_2$
S-(C)(H)
C-(S)$(C)(H)_2$

+ Correction between non neighboring atoms

$$\Delta_f H^\circ(298\,K) = \sum_i GAV_{\Delta_f H^\circ}(X_i)$$

$$S^\circ(298\,K) = \sum_i GAV_{S^\circ}(X_i) - Rln(\sigma/n_{opt})$$

$$C_P^\circ(T) = \sum_i GAV_{C_P^\circ}(X_i)$$

Source of Thermodynamic Data

Benson's Group Additivity Method

Uncertainties

- Benson's original data: ±2 kcal/mol for molecules, ±4 kcal/mol for radicals

New groups:

- Domalki and Hearing (1993)
- Cohen (1995)
- W. H. Green (2002)
- Bozzelli and Curran in THERM code

Codes based on Benson's group additivity

- Thergas
- THERM
- CanTherm
- CHETAH
- ……

New groups, new corrections and the accurate theoretical calculated group values make the uncertainties smaller!

Source of Thermodynamic Data

Statistical Thermodynamics

- *Partition function* (配分函数) measures the total number of possible energy states a molecule can be in at an equilibrium temperature

Total partition function $Q = Q_{trans} Q_{rot} Q_{vib} Q_{elec}$

Translational partition function $Q_{trans} = \left(\frac{2\pi M k_B T}{h^2}\right)^{\frac{3}{2}} V$

Rotational partition function $Q_{rot} = \frac{\pi^{\frac{1}{2}}}{\sigma} \left(\frac{8\pi^2 k_B T}{h^2}\right)^{\frac{3}{2}} \sqrt{I_a I_b I_c}$

Vibrational partition function $Q_{vib} = \prod_i e^{-h\nu_i/2kT} \cdot \left(1 - e^{-h\nu_i/k_B T}\right)^{-1}$

Electronic partition function $Q_{elec} = \sum_{i=1} g_i exp\left(-\frac{E_i}{k_B T}\right)$

Source of Thermodynamic Data

Quantum Chemistry Methods

Enthalpy and heat capability

- Total partition function to calculate enthalpy

$$S = Nk_B lnQ + Nk_B T \frac{dlnQ}{dT}$$

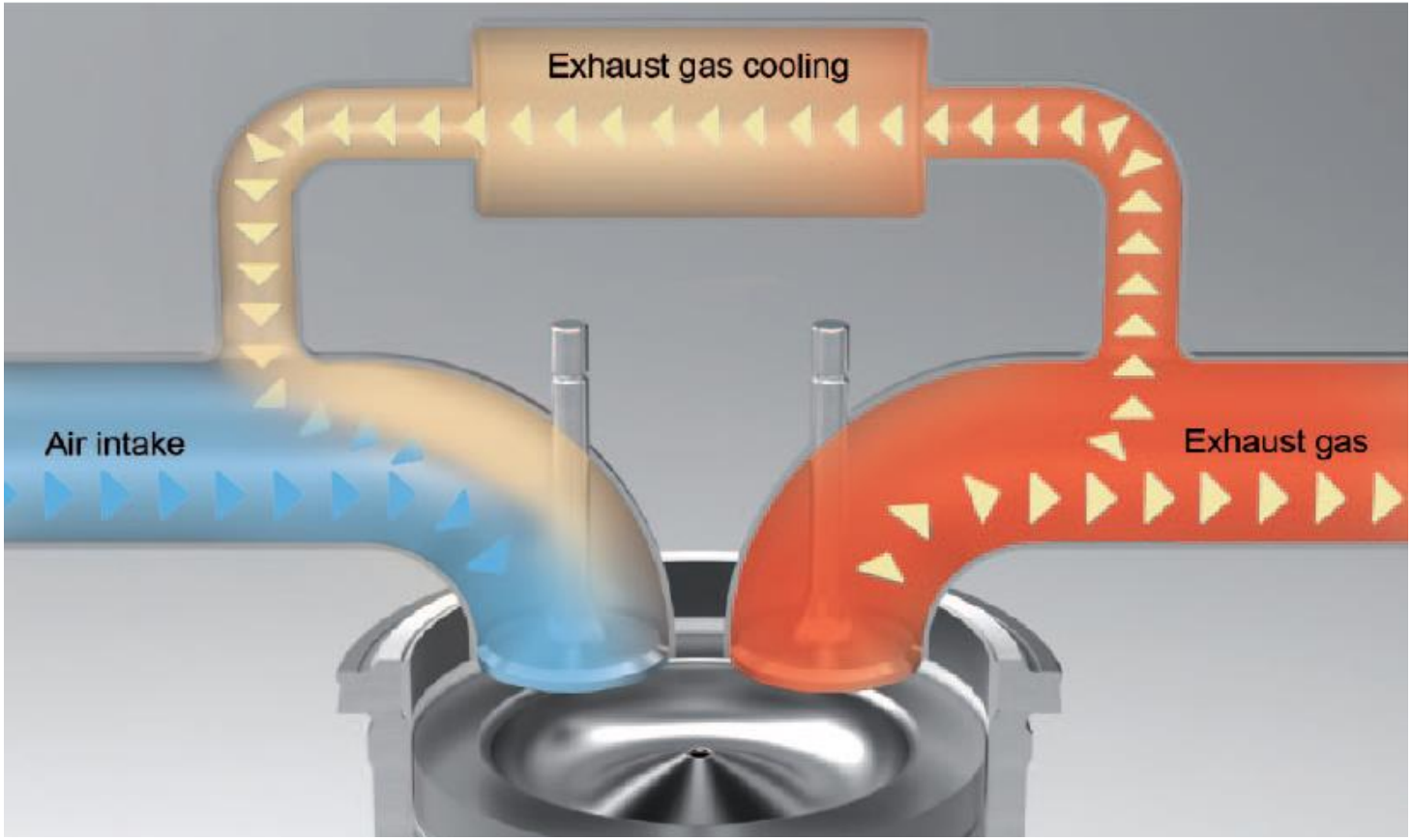
- Total partition function to calculate internal energy

$$U = Nk_B T^2 (\frac{\partial lnQ}{\partial T})_V$$

- Internal energy to calculate heat capability

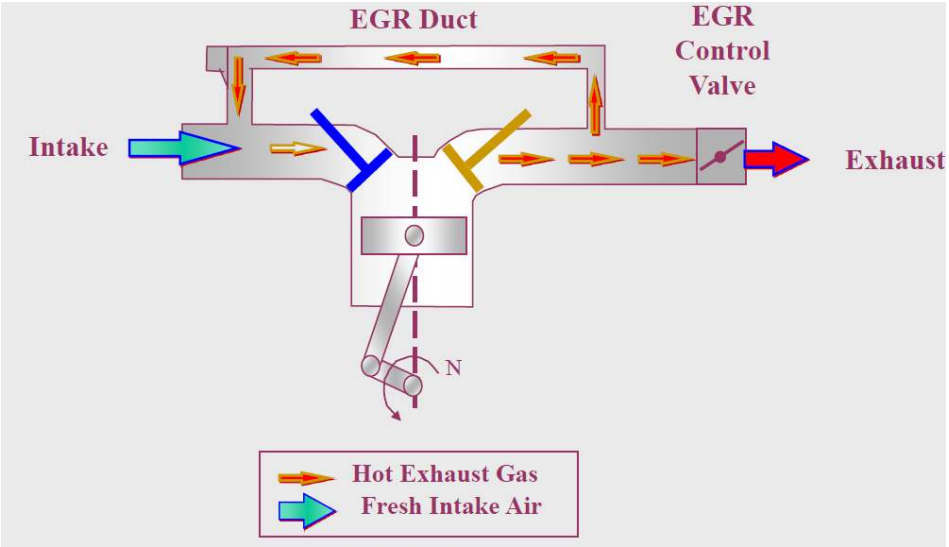
$$C_V = (\frac{\partial U}{\partial T})_V$$

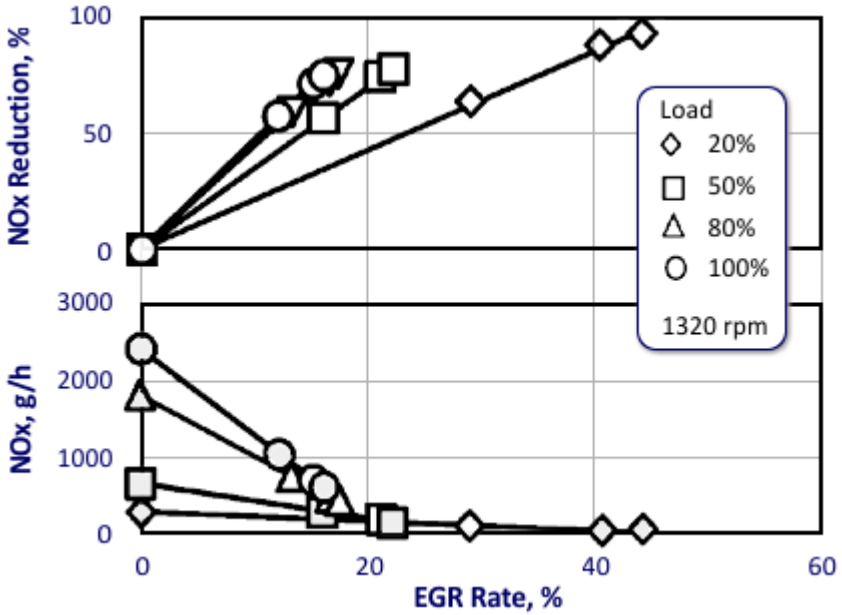
An Example of Application - EGR



- In exhaust gas recirculation (EGR), some of the exhaust gas is drawn off from the exhaust system, cooled and redirected back into the cylinders

An Example of Application - EGR

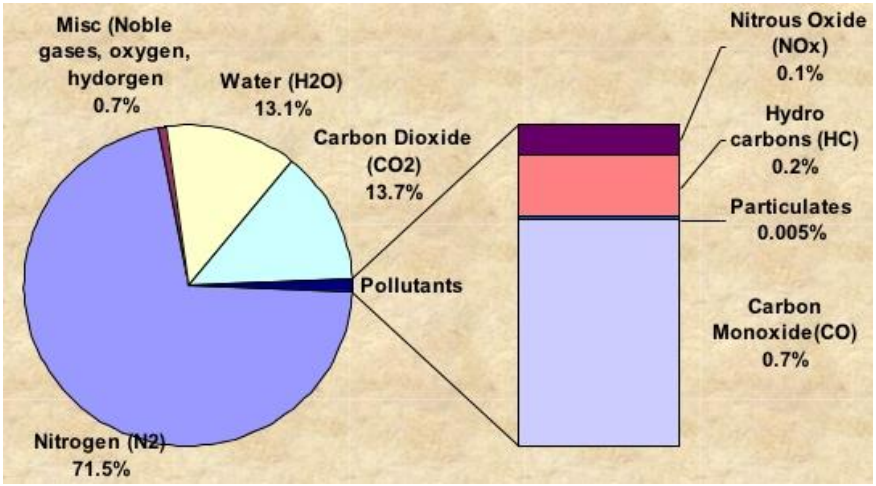


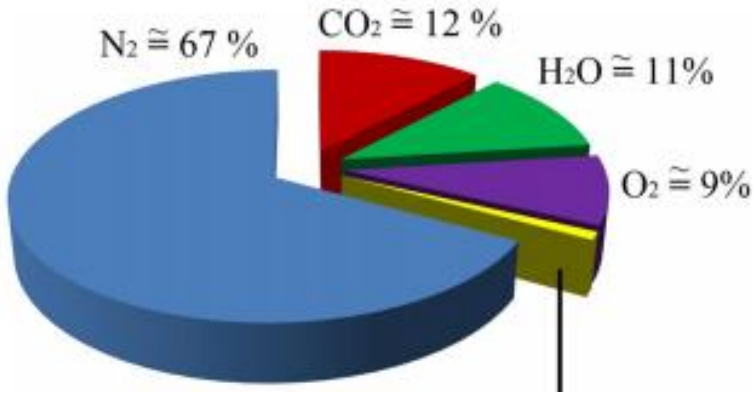


- Exhaust gas recirculation (EGR) is an effective strategy to control NOx emissions from diesel engines.
- The EGR reduces NOx through lowering the oxygen concentration in the combustion chamber, as well as through heat absorption.

An Example of Application - EGR

Gasoline engine exhaust



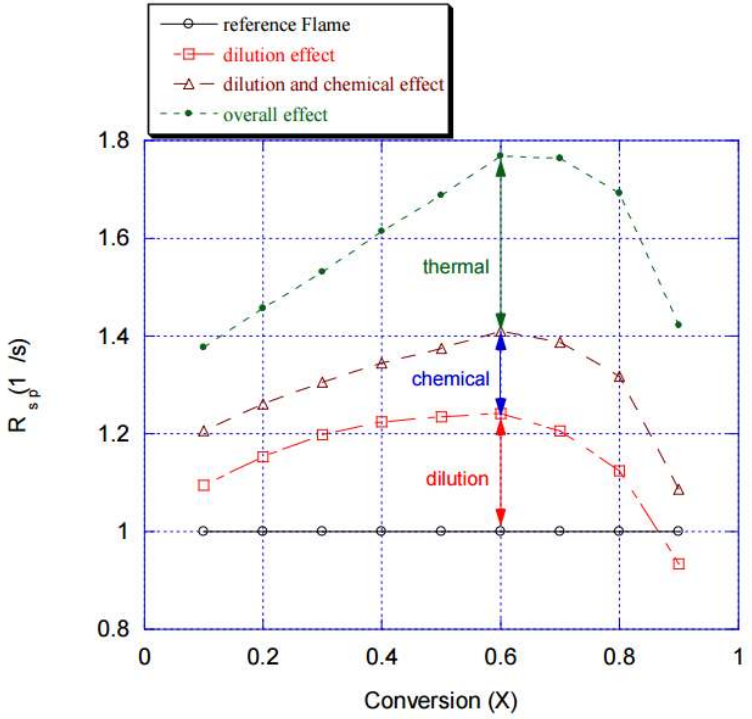


Diesel engine exhaust

- Diesel exhaust contains CO_2, H_2O, N_2 and O_2 in thermodynamically significant quantities and CO, UHC, NOx and soot in thermodynamically insignificant but environmentally harmful quantities.

An Example of Application - EGR

- *Dilution* – reduce reactant concentrations, reduce reaction rates
- *Thermal* – reduce flame temperature, reducing rate coefficients
- *Transport* – thermal/mass diffusivity (Lewis number) and Radiation
- *Chemical* – Reactions of "diluent" with fuel, oxidizer, and intermediates (e.g. $CO_2+H \rightarrow CO+OH$ and $HCO+M=H+CO$, $H_2O+O=2OH$)

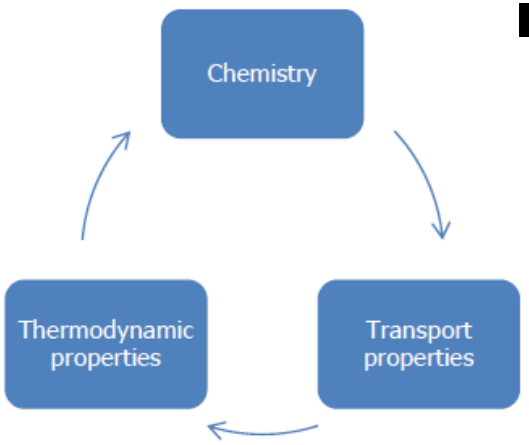


- *Thermal > Dilution > Chemical*

Outline of Chapter 2

- Introduction to Thermodynamics
- Fundamentals of Thermodynamics
- Thermodynamics in Combustion
- Transport Properties
- Transport in Combustion

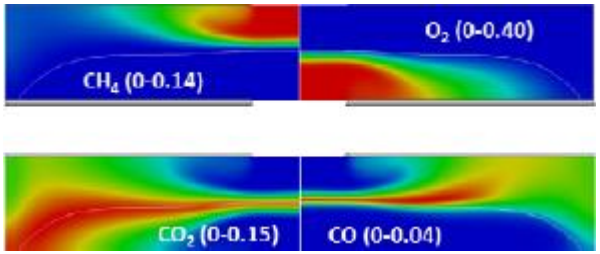
Transport in Combustion

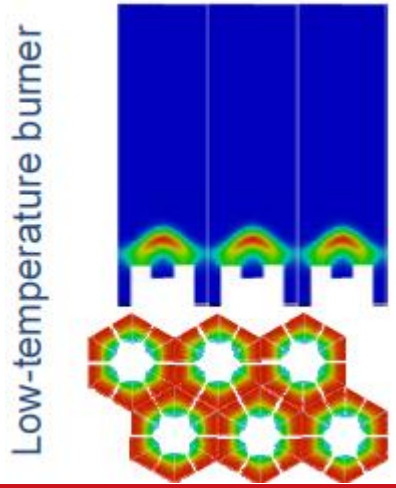


- Just as chemical reactions and their underlying parameters are essential for combustion modeling, molecular transport is important as well *for non-homogeneous combustion system*



- Viscosity
- Binary diffusion coefficients for species components
- Thermal conductivity
- Thermal diffusion





Transport Equations

Conservation Law

Continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla(\rho \mathbf{v}) = 0$$

Momentum equation

$$\rho \frac{\partial \mathbf{v}}{\partial t} + \rho \mathbf{v} \nabla \mathbf{v} = -\nabla p - [\nabla \cdot \boldsymbol{\tau}] + \rho \mathbf{g}$$

Viscous tensor (fluxes of momentum)

Energy equation

$$\rho C_P \frac{\partial T}{\partial t} + \rho C_P \mathbf{v} \nabla T = -\nabla \mathbf{q} - \left.\frac{\partial \ln \rho}{\partial \ln T}\right|_p \frac{Dp}{Dt} - \boldsymbol{\tau} : \nabla \mathbf{v}$$

Heat flux

Viscous dissipation term (usually negligible)

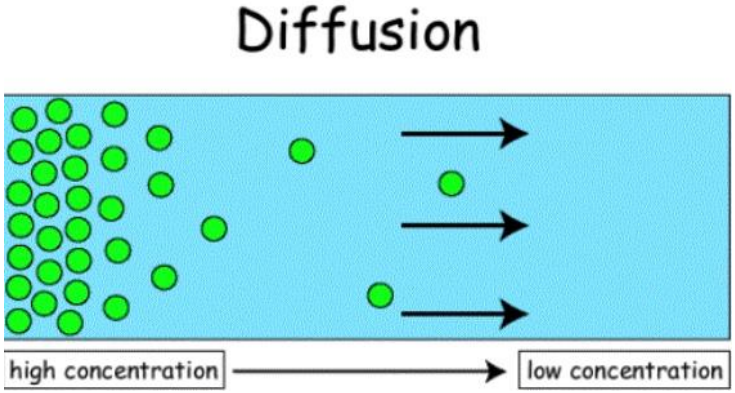
Species equations

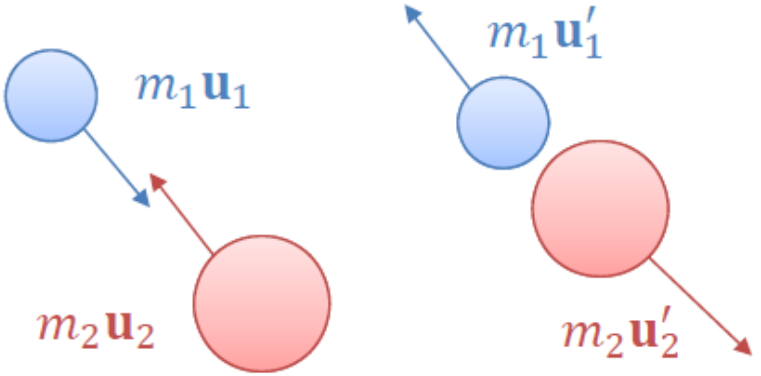
$$\rho \frac{\partial \omega_k}{\partial t} + \rho \mathbf{v} \nabla \omega_k = -\nabla \mathbf{j}_k + \dot{\Omega}_k$$

Mass flux

Simple Theory

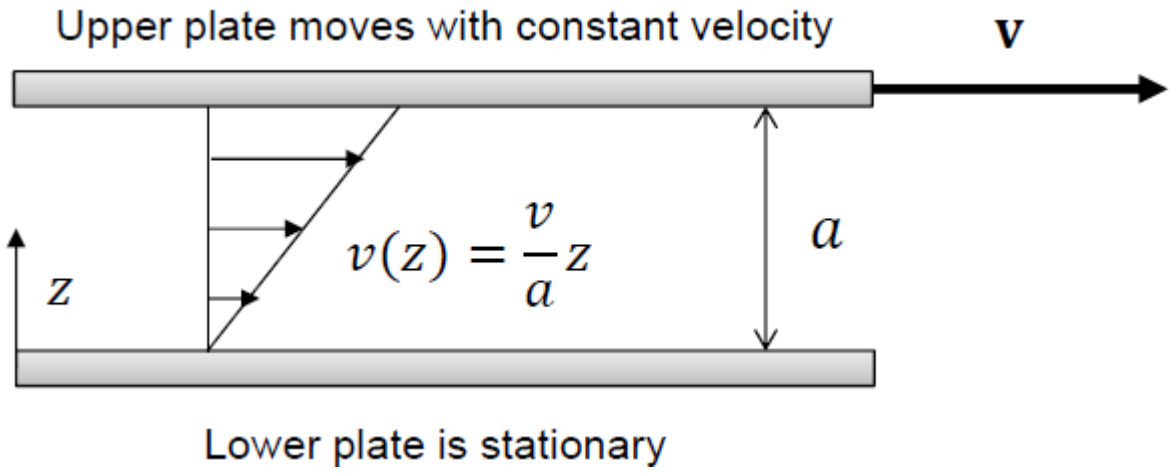
	Velocity gradient	Temperature gradient	Concentration gradient
Momentum	Newton's law		
Energy		Fourier's law	Dufour effect
Mass		Soret effect	Fick's law





Heat and mass transport

Simple Theory – Viscosity



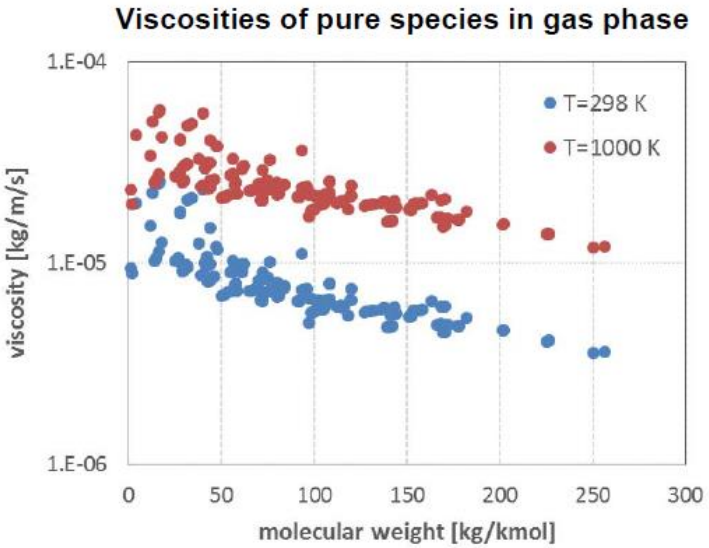
Newton's Law

Momentum flux

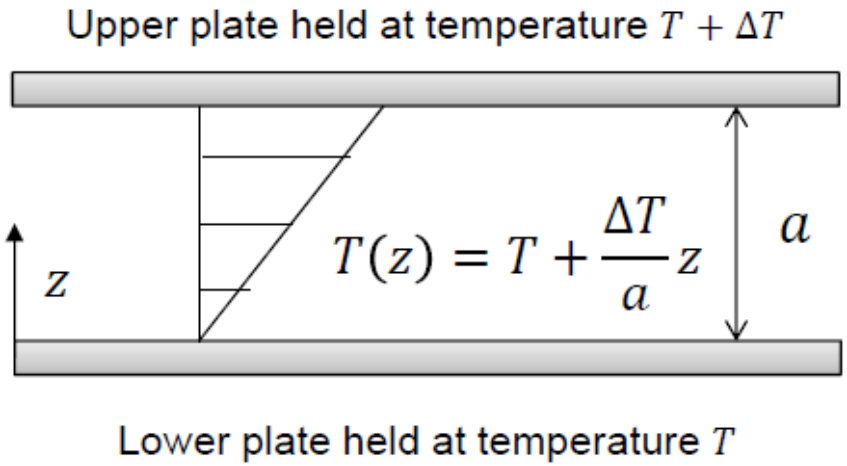
$$f = -\mu\frac{dv}{dz} = \mu\frac{v}{a}$$

Experimental observations

- Increases with the temperature as ~$T^{1/2}$
- Almost independent of pressure
- Decreases with the molecular weight



Simple Theory – Thermal Conductivity



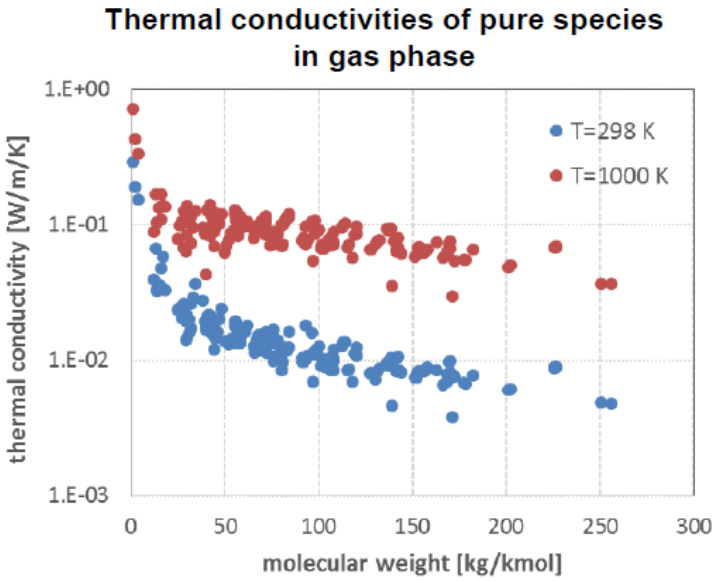
Fourier's Law

Heat flux

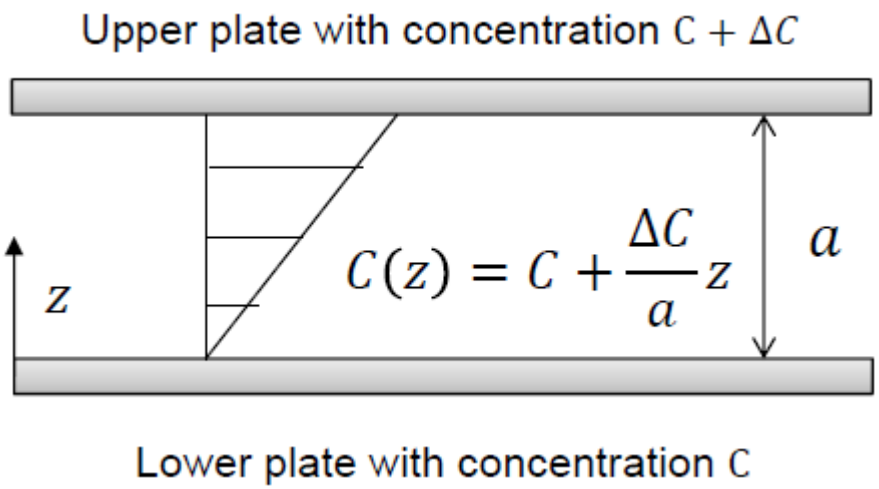
$$q = -\lambda \frac{dT}{dz} = -\lambda \frac{\Delta T}{a}$$

Experimental observations

- Increases with the temperature as ~$T^{1/2}$
- Almost independent of pressure
- Decreases with the molecular weight



Simple Theory – Mass Diffusivity



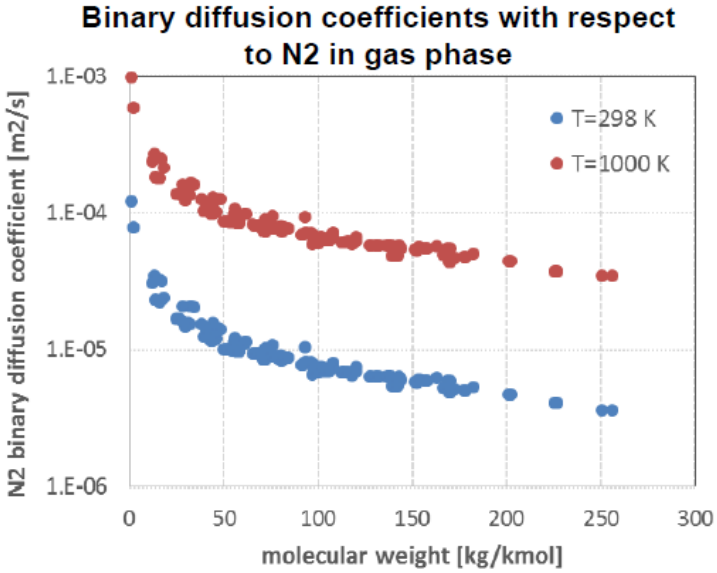
Fick's Law

Molar flux

$$J^* = -D\frac{dC}{dz} = -D\frac{\Delta C}{a}$$

Experimental observations

- Increases with the temperature as ~$T^{3/2}$
- Decreases with pressure as 1/p
- Decreases with the molecular weight



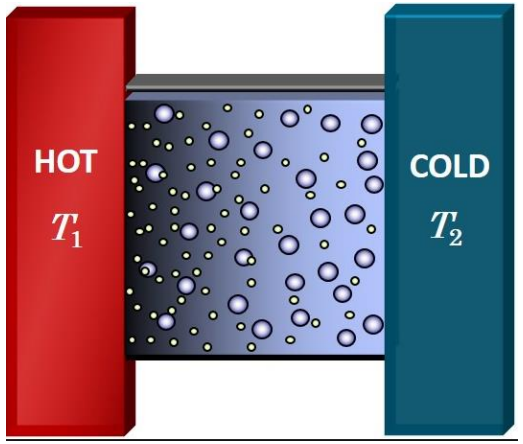
Simple Theory – Thermal Diffusion Coefficients

Soret effect (thermal diffusion)

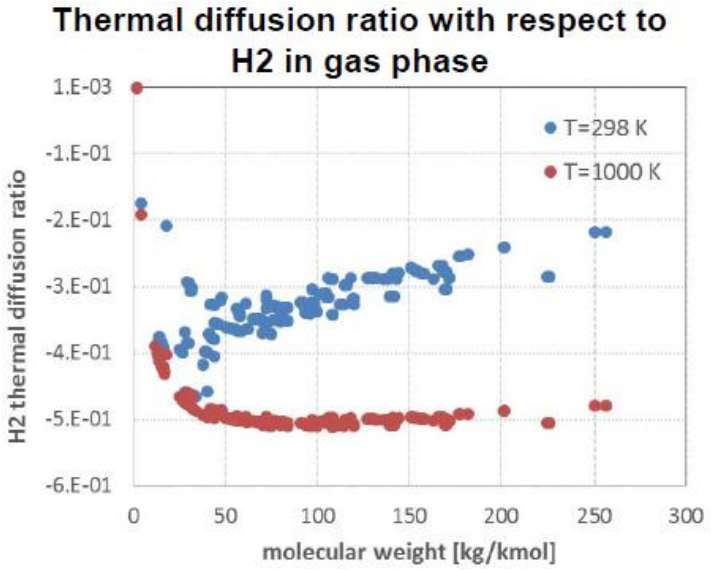
$$J^{th} = -\frac{D^T}{T}\nabla T = -D\theta\frac{1}{T}\nabla T$$

Fick's diffusion coefficient

Thermal diffusion ratio



- Thermal diffusion is important for light species such as H_2, H, He



Kinetic Theory of Gases

Number of molecules per unit volume

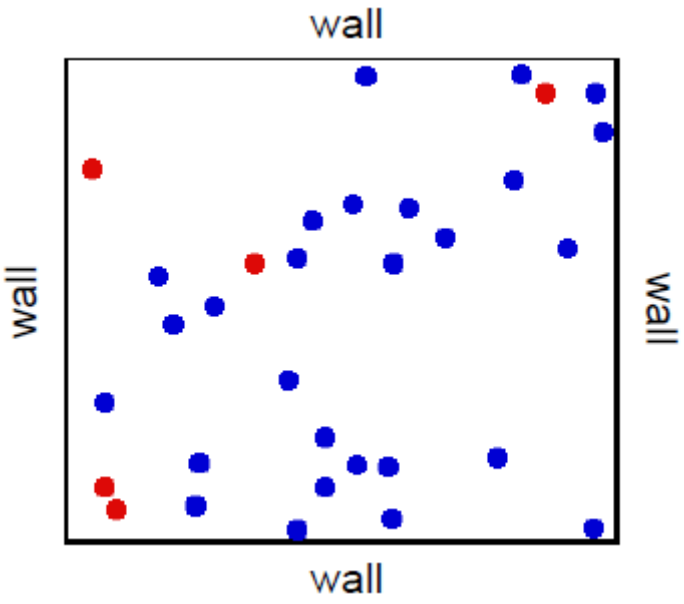
$$n_V = \frac{N}{V} = \frac{p}{k_B T}$$

Mean free path

$$l = \frac{1}{\sqrt{2}\pi d^2 n_V}$$

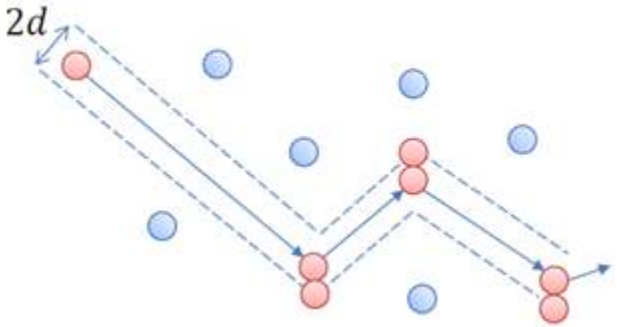
Mean speed

$$\bar{c} = \sqrt{\frac{8k_B T}{\pi m}}$$

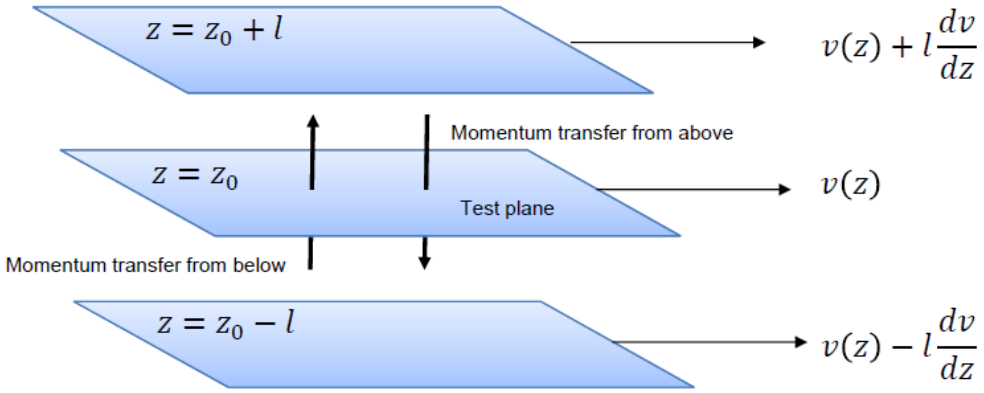


Number of molecules colliding with the wall per unit area and unit time

$$Z = \frac{\bar{c}}{4} n_V = n_V \sqrt{\frac{k_B T}{2\pi m}}$$



Kinetic Theory of Gases – Viscosity



Net time rate of change of momentum:

$$M = Zm\left\{\left[v(z) - l\left(\frac{du}{dz}\right)\right] - \left[v(z) + l\left(\frac{du}{dz}\right)\right]\right\} = -2Zml\frac{dv}{dz}$$

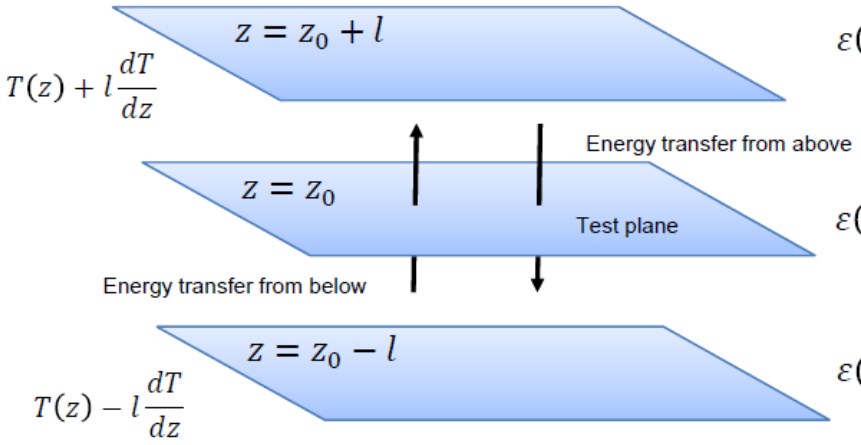
$$\begin{cases} f = M = -2Zml\dfrac{du}{dz} \\ f = -\mu\dfrac{du}{dz} \end{cases}$$

Newton's Law

$$\mu = 2Zml = \frac{\sqrt{\pi m k_B T}}{\pi^2 d^2}$$

- Independent of pressure
- Increases with the square root of T

Kinetic Theory of Gases – Thermal Conductivity



$$\varepsilon(z+l) = \varepsilon(z) + l\frac{C_V}{N_{AV}}\frac{dT}{dz}$$

$$\varepsilon(z-l) = \varepsilon(z) - l\frac{C_V}{N_{AV}}\frac{dT}{dz}$$

Net heat flux:

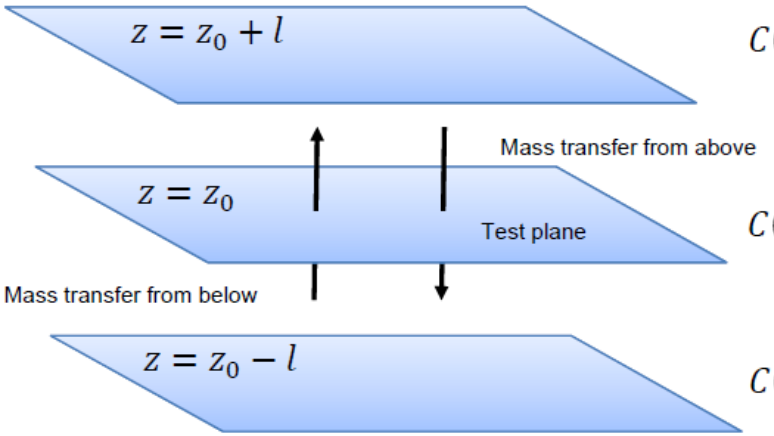
$$q = Z\left\{\left[\varepsilon(z) - l\frac{C_V}{N_{AV}}\left(\frac{dT}{dz}\right)\right] - \left[\varepsilon(z) + l\frac{C_V}{N_{AV}}\left(\frac{dT}{dz}\right)\right]\right\} = -2Zl\frac{C_V}{N_{AV}}\frac{dT}{dz}$$

$$\begin{cases} q = -2Zl\dfrac{C_V}{N_{AV}}\dfrac{dT}{dz} = \dfrac{\bar{c}}{2}\dfrac{n_V}{N_{AV}}lC_V\dfrac{dT}{dz} \\ q = -\lambda\dfrac{dT}{dz} \end{cases} \Rightarrow \lambda = \frac{\bar{c}}{2}\frac{n_V}{N_{AV}}lC_V = \frac{1}{\pi\sqrt{\pi}}\sqrt{\frac{k_BT}{m}}\frac{C_v}{N_{AV}d^2}$$

Fourier's Law

- **Independent of pressure**
- **Increases with the square root of T**

Kinetic Theory of Gases – Mass Diffusivity



$$C(z+l) = C(z) + l\frac{dC}{dz}$$

$C(z)$ Concentration of species

$$C(z-l) = C(z) - l\frac{dC}{dz}$$

Net molecular flux:

$$J^* = \frac{Z}{n_V}\left\{\left[C(z) - l\left(\frac{dC}{dz}\right)\right] - \left[C(z) + l\left(\frac{dC}{dz}\right)\right]\right\} = -2\frac{Z}{n_V}l\frac{dC}{dz}$$

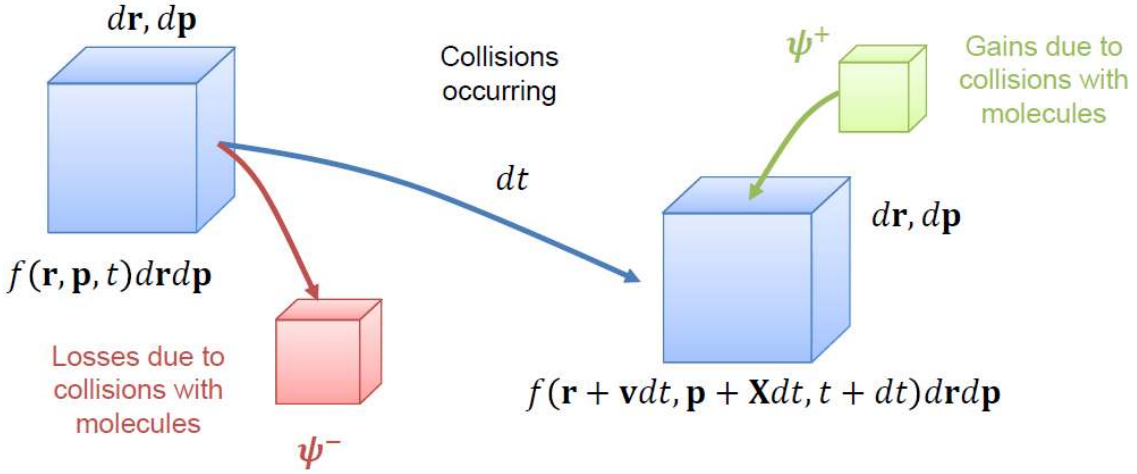
$$\begin{cases} J^* = 2\dfrac{Z}{n_V}l\dfrac{dC}{dz} = \dfrac{\bar{c}}{2}l\dfrac{dC}{dz} \\ J^* = -D\dfrac{dC}{dz} \end{cases}$$

$$D = \frac{\bar{c}}{2}l = \frac{1}{\pi\sqrt{\pi}}\sqrt{\frac{k_B^3T^3}{m}}\frac{1}{pd^2}$$

Fick's Law

- Decreases with pressure as 1/p
- Increases with the temperature as $T^{3/2}$

Rigorous Theory – Boltzmann Equation



- **The Boltzmann equation is an integro-differential equation describing the time evolution of t.**

Boltzmann Equation

$$\frac{\partial f}{\partial t} + \frac{\mathbf{p}}{m}\frac{\partial f}{\partial \mathbf{r}} + \mathbf{X}\frac{\partial f}{\partial \mathbf{p}} = [\psi^+ - \psi^-]$$

The loss and gain terms contain contributions from all possible collisions that can deflect the molecules during the time step dt.

They are described through proper collision integrals.

Rigorous Theory – Boltzmann Equation

From molecular distribution

$$\begin{cases} \mathbf{q} = \int f \dfrac{mV^2}{2} V_i d\mathbf{v} \\ \mathbf{q} = -\lambda \nabla T \end{cases}$$

Fourier's law

Chapman-Enskog solution of the Boltzmann equation

$$\lambda = \frac{75}{64}\left(\frac{k_B^3 T \pi}{m}\right)^{1/2} \frac{1}{\Omega^{(2,2)}}$$

From molecular distribution

$$\begin{cases} T_{ij} = \int f m \mathbf{V}\mathbf{V} d\mathbf{v} \\ T_{ij} = -\mu\left(\dfrac{\partial v_i}{\partial x_j} + \dfrac{\partial v_j}{\partial x_i}\right) \end{cases}$$

Newton's law

Chapman-Enskog solution of the Boltzmann equation

$$\mu = \frac{5}{16}(m\pi k_B T)^{1/2} \frac{1}{\Omega^{(2,2)}}$$

Collision integrals Ω: the contributions from all possible collisions can deflect molecules

Reduced relative velocity

$$\Omega_{ij}^{(l,s)} = \left(\frac{2\pi k_B T}{m_{ij}}\right)^{1/2} \int\int e^{-\gamma_{ij}^2} \gamma_{ij}^{2s+3} \left(1 - \cos^l \chi\right) b db d\gamma_{ij}$$

Intermolecular potential

deflection angle

$$\chi(b.\gamma_{ij}) = \pi - 2b \int_{r_m}^{\infty} \frac{1}{r^2} \left(1 - \frac{b^2}{r^2} - \frac{u_{ij}(r)}{k_B T \gamma_{ij}^2}\right)^{-1/2} dr$$

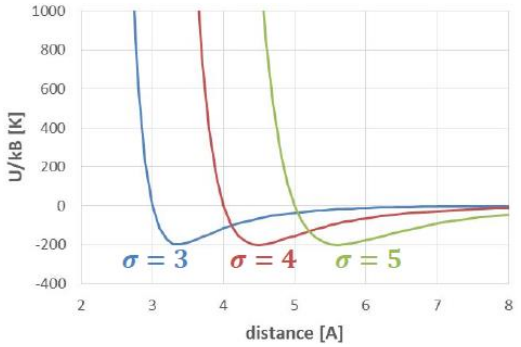
Rigorous Theory – Intermolecular Potential

Lennard-jones potential (For monoatomic molecules)

$$U(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

- ε: net collision diameter
- σ: well-depth

Species	$\varepsilon/k_B\ (K)$	$\sigma\ (\dot{A})$
Ar	136.5	3.330
O2	107.4	3.458
N2	97.53	3.621
CO2	244.0	3.763
CH4	141.4	3.746
H2O	572.4	2.605



Stockmayer potential (For diatomic and polyatomic molecules)

$$U(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} - \delta^{*}\left(\frac{\sigma}{r}\right)^{3}\right]$$

$$\delta^{*} = \frac{1}{4}\frac{\bar{\mu}\bar{\mu}}{\varepsilon\sigma^{3}}\chi$$

$\bar{\mu}$ Dipole moment

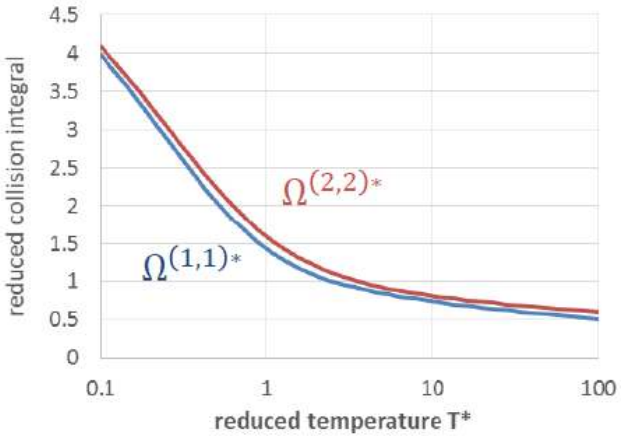
- The additional term accounts for the interaction arising from permanent, non uniform electric densities within each molecules

- Geometric function depending on the orientation of the molecules
- In the case where one or both the dipole moments are zero, the Stockmayer potential reduces to the Lennard-Jones potential

Rigorous Theory – Reduced Collision Integrals

Lennard-Jones potential

$$\begin{cases}\Omega^{(1,1)*}(T^*) \approx a_1 T^{*-a_2} + (T^* + a_3)^{-a_4} \\ \Omega^{(2,2)*}(T^*) \approx b_1 T^{*-b_2} + (T^* + b_3)^{-b_4}\end{cases}$$



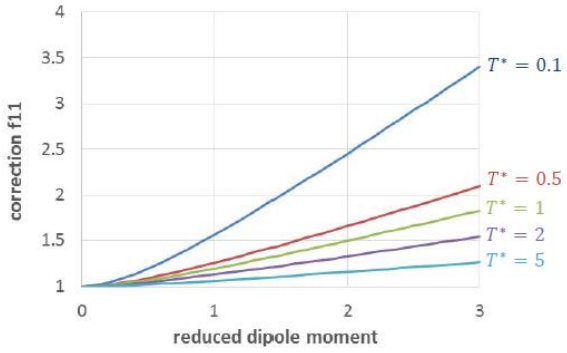
Stockmayer potential

$$\begin{cases}\Omega^{(1,1)*}(T^*) \approx [a_1 T^{*-a_2} + (T^* + a_3)^{-a_4}] f^{(1,1)} \\ \Omega^{(2,2)*}(T^*) \approx [b_1 T^{*-b_2} + (T^* + b_3)^{-b_4}] f^{(2,2)}\end{cases}$$

$$\begin{cases} f^{(1,1)}(T^*, \widetilde{\delta^*}) = 1 + \dfrac{\left(e^{a_5/T^*} - e^{-a_6/T^*}\right)\left(\widetilde{\delta^*}\right)^2}{2 + 2.5\widetilde{\delta^*}} \\ f^{(2,2)}(T^*, \widetilde{\delta^*}) = 1 + \dfrac{\left(e^{b_5/T^*} - e^{-b_6/T^*}\right)\left(\widetilde{\delta^*}\right)^2}{2 + 2.5\widetilde{\delta^*}} \end{cases}$$

$$\widetilde{\delta^*} = 2\delta^*$$

Correction factors



Geometric in real gases the cross-section depends on the temperature. The reduced collision integral accounts for these dependencies.

Theory Summary

	Simple theory	Rigorous theory (Hard spheres)	Rigorous theory (general case)
Viscosity	$\mu = \frac{1}{\pi}\frac{\sqrt{\pi m k_B T}}{\pi\sigma^2}$	$\mu = \frac{5}{16}\frac{\sqrt{\pi m k_B T}}{\pi\sigma^2}$	$\mu = \frac{5}{16}\frac{\sqrt{\pi m k_B T}}{\pi\sigma^2}\frac{1}{\Omega^{(2,2)*}}$
Thermal conductivity	$\lambda = \frac{1}{\pi}\sqrt{\frac{k_B T}{\pi m}}\frac{C_v}{N_{AV}\sigma^2}$	$\lambda = \frac{25}{32}\sqrt{\frac{k_B T}{\pi m}}\frac{C_v}{N_{AV}\sigma^2}$	$\lambda = \frac{25}{32}\sqrt{\frac{k_B T}{\pi m}}\frac{C_v}{N_{AV}\sigma^2}\frac{1}{\Omega^{(2,2)*}}$
Mass diffusivity	$\Gamma = \frac{1}{\pi}\sqrt{\frac{\pi k_B^3 T^3}{m}}\frac{1}{\pi p\sigma^2}$	$\Gamma = \frac{3}{8}\sqrt{\frac{\pi k_B^3 T^3}{m}}\frac{1}{\pi p\sigma^2}$	$\Gamma = \frac{3}{8}\sqrt{\frac{\pi k_B^3 T^3}{m}}\frac{1}{\pi p\sigma^2}\frac{1}{\Omega^{(1,1)*}}$

- **The rigorous theory for hard sphere molecules gives the same general form of the transport properties evaluated through the simple model (i.e. the dependencies of T, p and m are exactly the same). Only the numerical coefficients are different (because of non-rigorous assumptions in the simple model).**

Mass Diffusion – Binary Diffusion Coefficients

Convective and diffusive velocities

Species equations

Mass flux

$$\rho\frac{\partial \omega_k}{\partial t} + \rho\mathbf{v}\nabla\omega_k = -\nabla\mathbf{j}_k + \dot{\Omega}_k$$

Conservation of momentum after collision

$$m_1\mathbf{u}_1 + m_2\mathbf{u}_2 = m_1\mathbf{u}_1' + m_2 m_1\mathbf{u}_2'$$

Stefan-Maxwell equation for binary mixtures and Fick's law for binary mixtures

$$\nabla X_k = \sum_{j=1}^{NC}\frac{X_k X_j}{\Gamma_{kj}}(\mathbf{V}_j - \mathbf{V}_k) + (\omega_k - X_k)\frac{\nabla p}{p} + \sum_{j=1}^{NC}\frac{X_k X_j}{\rho\Gamma_{kj}}\left(\frac{\theta_j}{\omega_j} - \frac{\theta_k}{\omega_k}\right)\frac{\nabla T}{T}$$

The constraint equation:

$$\sum_{k=1}^{NC}\rho\omega_k\mathbf{V}_k = 0$$

Binary diffusion coefficients:

$$\Gamma_{jk} = \frac{3}{16}\sqrt{\frac{\pi k_B^3 T^3}{m_{jk}}}\frac{1}{p\pi\sigma_{jk}^2\Omega_{jk}^{(1,1)*}}$$

Mass Diffusion – Mixture-Averaged

- Approximate methods to determine transport properties in a mixture from the pure species values and binary diffusion coefficients
- Compromise between accuracy and computational expense

$$\mu = \sum_{k=1}^{K} \frac{X_k \mu_k}{\sum_{j=1}^{K} X_j \Phi_{kj}}$$

The Wilke formula for mixture viscosity

$$\Phi_{kj} = \frac{1}{\sqrt{8}} \left(1 + \frac{W_k}{W_j}\right)^{-1/2} \left(1 + \left(\frac{\mu_k}{\mu_j}\right)^{1/2} \left(\frac{W_j}{W_k}\right)^{1/4}\right)^2$$

μ_k: the pure species k viscosity

$$\lambda = \frac{1}{2}\left(\sum_{k=1}^{K} X_k \lambda_k + \frac{1}{\sum_{k=1}^{K} X_k / \lambda_k}\right)$$

Mixture-averaged thermal conductivity

$$D'_{km} = \frac{1 - Y_k}{\sum_{j \neq k}^{K} X_j / \mathcal{D}_{kj}}$$

Mixture-averaged diffusion coefficients

- No reliable mixture-averaged theory is available for computing the thermal diffusion coefficients

Mass Diffusion – Multicomponent

- The Chapman-Enskog theory provides the basis for multicomponent transport properties. The thermal conductivities and the thermal diffusion coefficients can be calculated from the solution of a system of linear equations defined by the L matrix.

Nine-block submatrices

$$\begin{bmatrix} \mathbf{L}^{00,00} & \mathbf{L}^{00,10} & \mathbf{0} \\ \mathbf{L}^{10,00} & \mathbf{L}^{10,10} & \mathbf{L}^{10,01} \\ \mathbf{0} & \mathbf{L}^{01,10} & \mathbf{L}^{01,01} \end{bmatrix} \begin{bmatrix} \mathbf{a}^1_{00} \\ \mathbf{a}^1_{10} \\ \mathbf{a}^1_{01} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{X} \\ \mathbf{X} \end{bmatrix}$$

Vector of mole fractions

Thermal conductivity **Thermal coefficients** **Thermal diffusion ratios** **Thermal diffusion velocity**

$$\lambda = \lambda_{trans} + \lambda_{int} \qquad \theta_k = \frac{8 m_k X_k}{5R} a^1_{00,k} \Rightarrow \theta_k = \sum_{j \neq k}^{NC} \theta_{kj} \qquad \mathbf{V}_k^T = -\frac{D_{k,mix}\theta_k}{X_k}\frac{1}{T}\nabla T$$

$$\lambda_{trans} = -4 \sum_{k=1}^{NC} X_k \, a^1_{10,k}$$

$$\theta_{jk} = \frac{15}{2} \frac{(2A^*_{jk}+5)(6C^*_{jk}-5)}{A^*_{jk}(16A^*_{jk}-12B^*_{jk}+55)} \frac{MW_j - MW_k}{MW_j + MW_k} X_j X_k$$

$$\lambda_{int} = -4 \sum_{k=1}^{NC} X_k \, a^1_{01,k}$$

- When thermal diffusion is important, the rigorous multicomponent theory should be used.

Outline of Chapter 2

- Introduction to Thermodynamics
- Fundamentals of Thermodynamics
- Thermodynamics in Combustion
- Transport Properties
- Transport in Combustion

Transport Data for Combustion Modeling

Example of file containing the transport data for each species in TRANLIB format

Species	Geometry	ε/k_B	σ	μ	α	Z_{rot}
H2	1	38.000	2.920	0.000	0.790	280.000
HE	0	10.200	2.576	0.000	0.000	0.000
C6H6	2	468.500	5.230	0.000	10.30	1.000
H2O	2	572.400	2.605	1.844	0.000	4.000

Name of the species (must be the same reported in the thermodynamic and kinetic files)

An index indicating the geometrical configuration. If the index is 0, the molecule is a single atom. If the index is 1 it is linear, and if it is 2, the molecule is nonlinear.

The potential well depth (in K).

The collision diameter (in angstroms)

The dipole moment (in Debye)

The polarizability (in cubic angstroms)

The rotational relaxation collision number at 298 K

Calculation of Transport Properties

1. Read the transport parameters from the TRANLIB file

2. Calculate the reduced masses and the combined collision integrals, potential well-depths and reduced dipole moments for all the possible pairs

3. Evaluate the collision integrals $\Omega^{(2,2)*}$ for all the pure species and the collisions integral $\Omega^{(1,1)*}$ for all the possible pairs

4. Calculate the pure species viscosities and thermal conductivitis for all the species and the binary diffusion coefficients for all the possible pairs

5. Apply the proper mixture average rules to calculate the mixture thermal conductivity and viscosity
Apply the proper formulation (Stefan-Maxwell vs Fick) to calculate the diffusion velocities

Homework

Estimate the adiabatic flame temperature of a constant pressure reactor burning a stoichiometric mixture of H_2 and air at 101.3 kPa and 25 °C at the inlet. *(give a detailed calculation process)*

Thanks for your attention



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